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Chava et al.

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(54) **PACKAGE COMPRISING INTEGRATED DEVICES AND BRIDGE COUPLING TOP SIDES OF INTEGRATED DEVICES**

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H01L 23/00 (2006.01)

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(52) **U.S. Cl.**

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(58) **Field of Classification Search**

CPC H01L 23/5385; H01L 24/16; H01L 24/81

See application file for complete search history.

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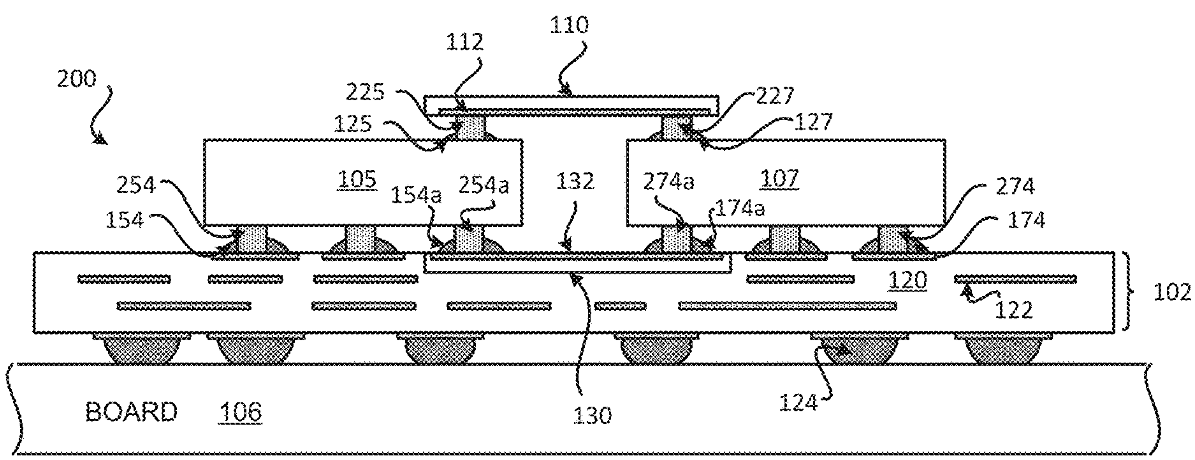
(74) *Attorney, Agent, or Firm* — Loza & Loza, LLP

(57) **ABSTRACT**

A package comprising a substrate; a first integrated device coupled to the substrate through at least a first plurality of solder interconnects; a second integrated device coupled to the substrate through at least a second plurality of solder interconnects; a first bridge coupled to the first integrated device and the second integrated device through at least a third plurality of solder interconnects, wherein the first bridge is configured to provide at least one first electrical path between the first integrated device and the second integrated device, and wherein the first bridge is coupled to a top portion of the first integrated device and a top portion of the second integrated device, through at least the third plurality of solder interconnects; and a second bridge coupled to the first integrated device and the second integrated device through a fourth plurality of solder interconnects.

25 Claims, 16 Drawing Sheets



(51) **Int. Cl.**
H01L 25/00 (2006.01)
H01L 25/10 (2006.01)

(52) **U.S. Cl.**
CPC ***H01L 25/105*** (2013.01); ***H01L 25/50*** (2013.01); *H01L 2224/16227* (2013.01)

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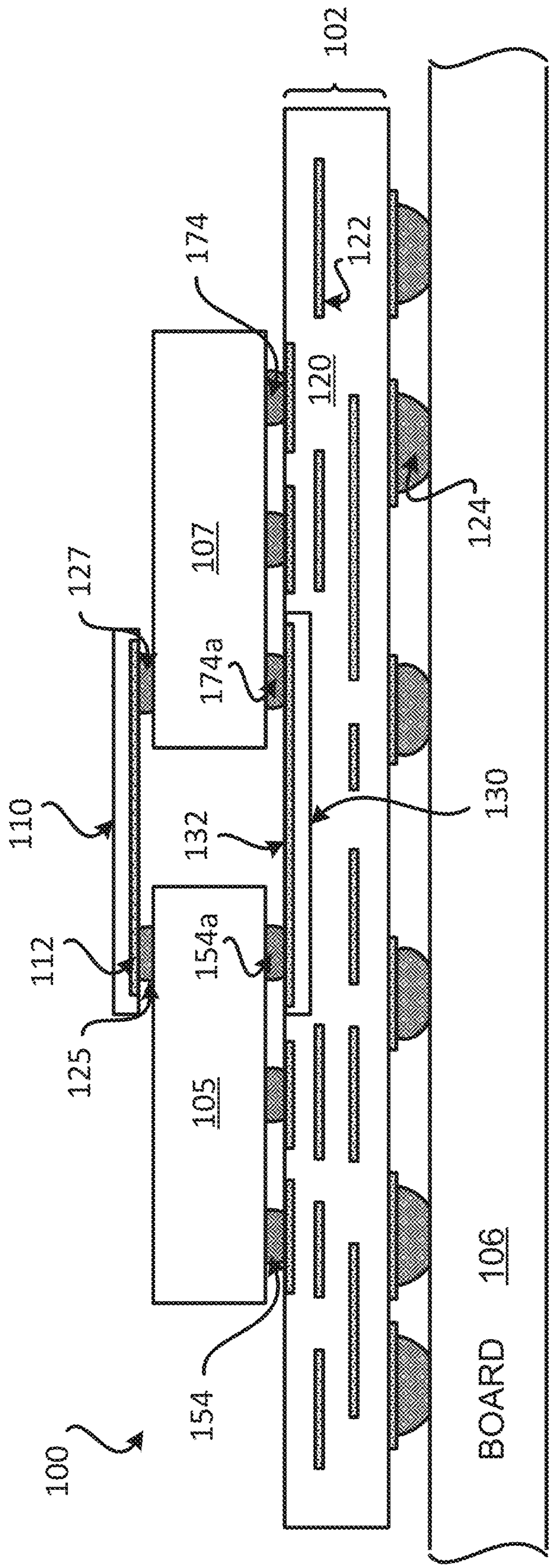


100
102
105
106
107
110
112
120
122
124
125
127
130
132
154
154a
174
174a
BOARD

FIG. 1

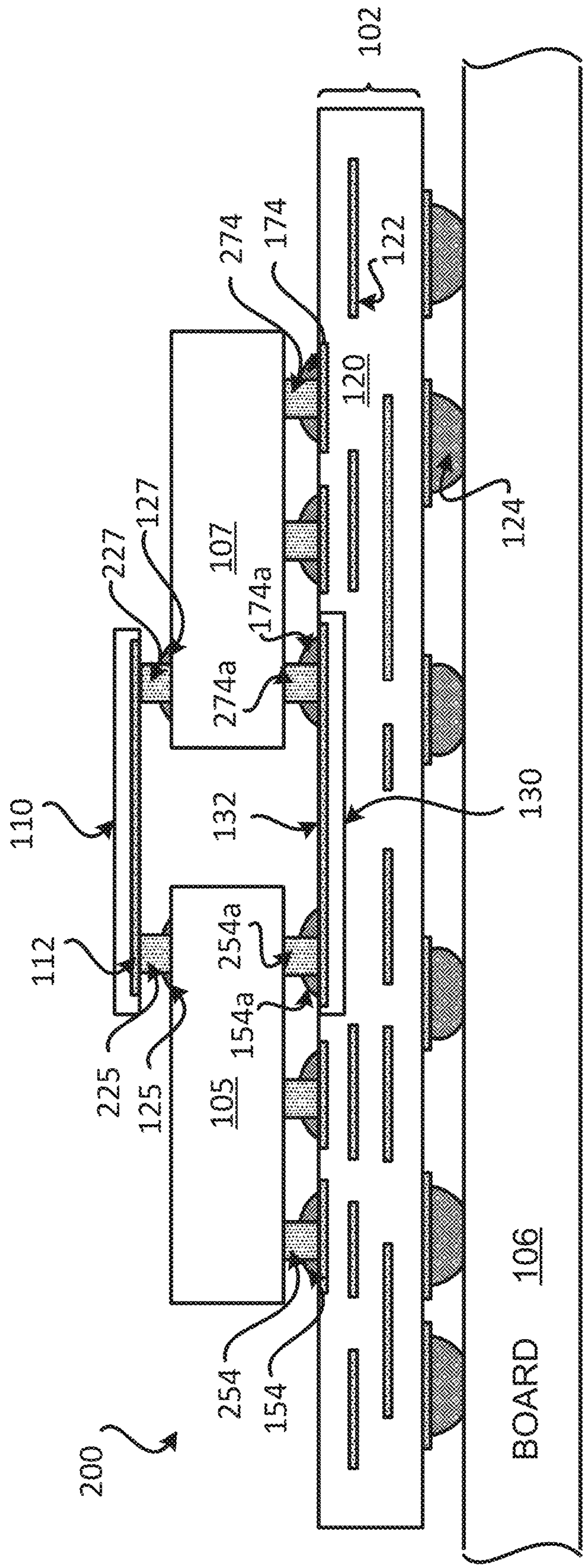

102
274
174
122
120
124
127
227
107
174a
274a
110
132
130
112
254a
154a
225
125
105
106
BOARD
254
154
200

FIG. 2

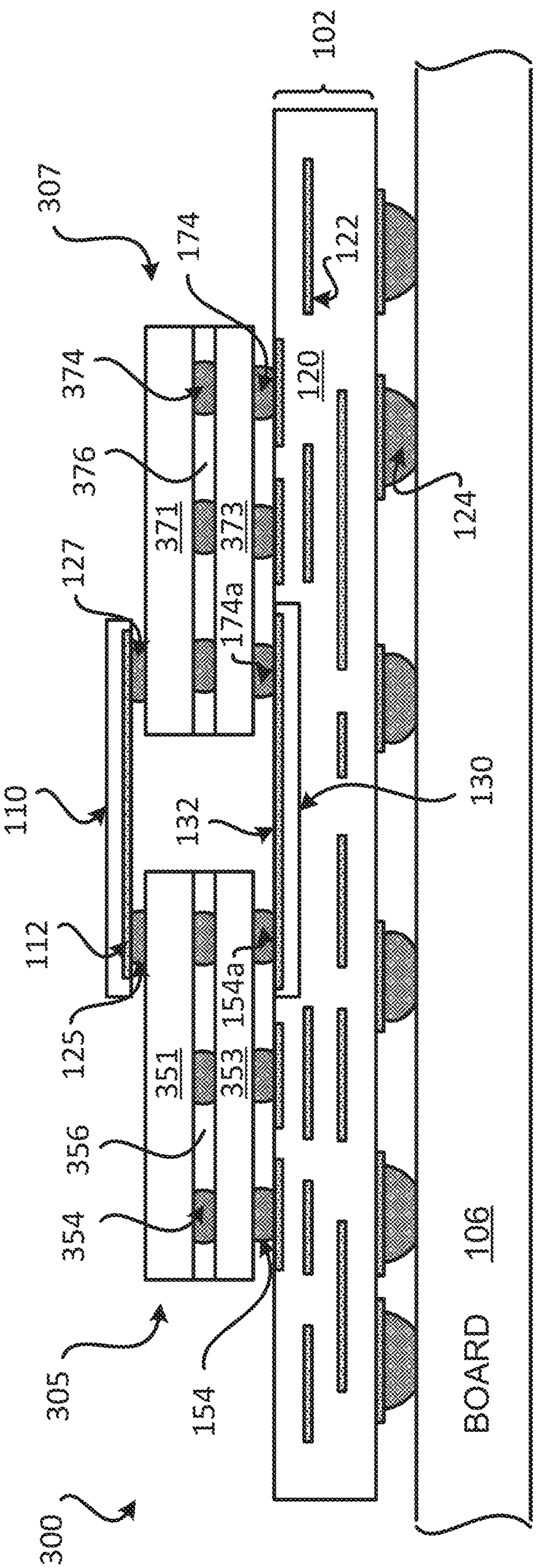
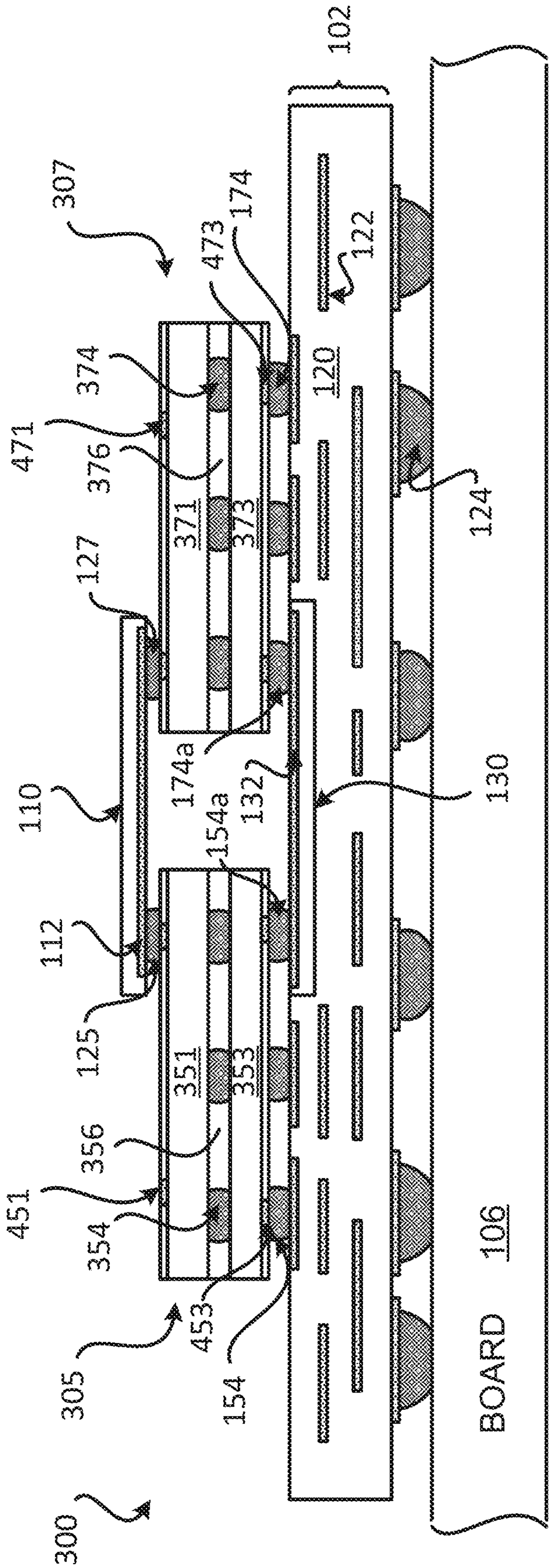

102
307
174
122
374
120
376
124
371
373
127
174a
110
132
130
112
125
154a
351
353
356
354
106
BOARD
305
154
300

FIG. 3



300
305
307
102
106
BOARD
110
112
120
122
124
125
127
130
132
154
154a
174
174a
351
353
354
356
371
373
374
376
451
453
471
473

FIG. 4

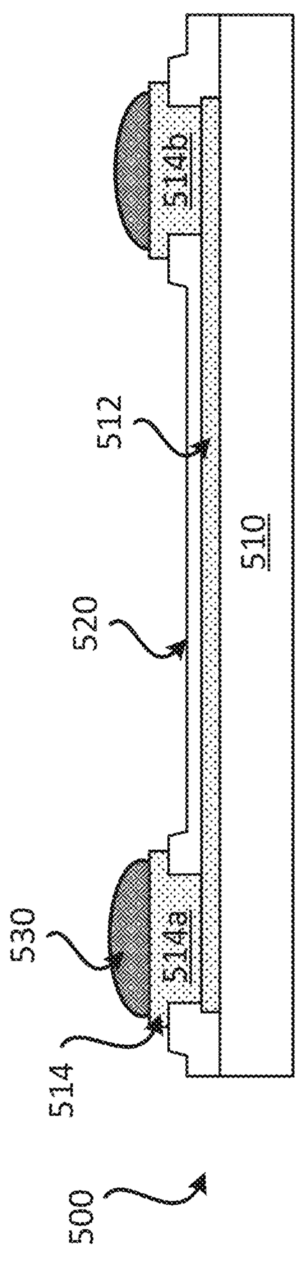


FIG. 5

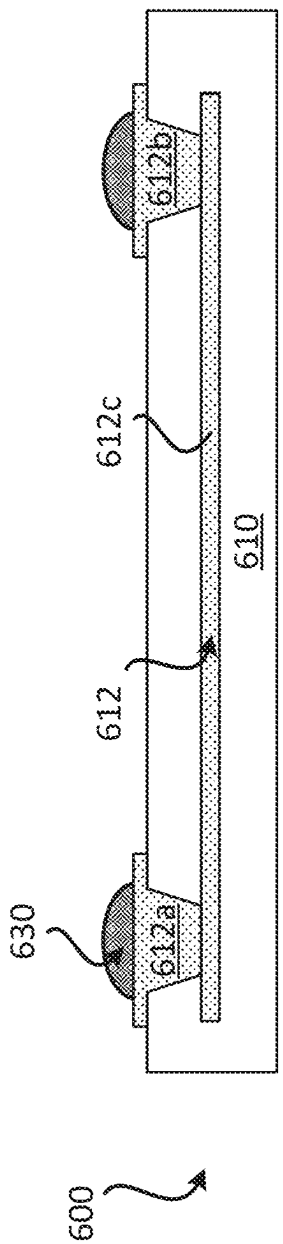
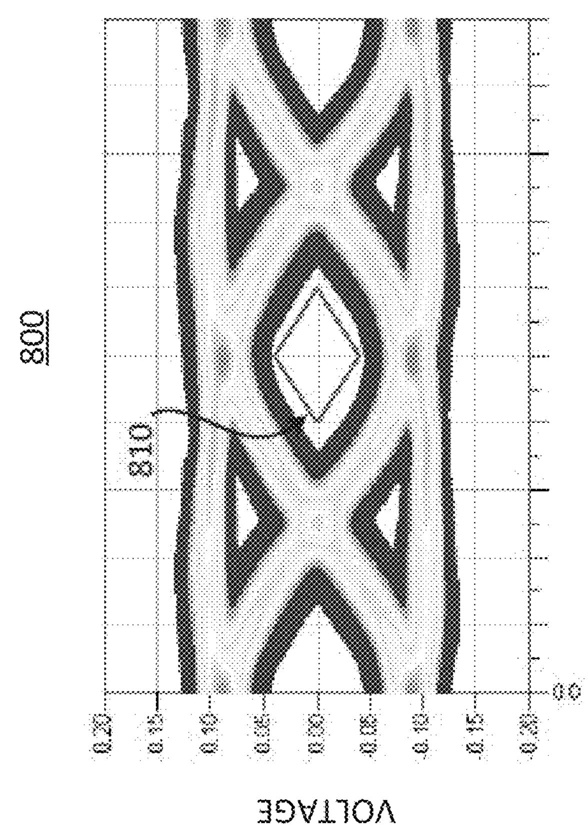
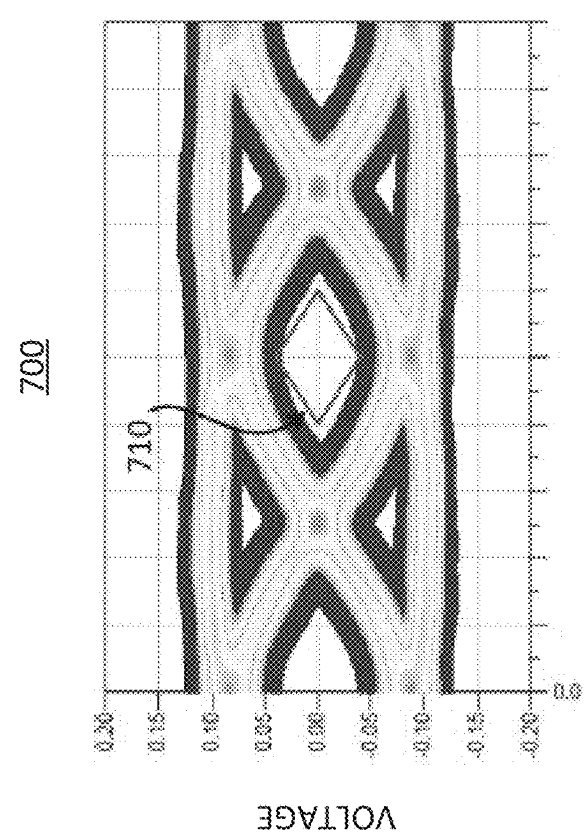


FIG. 6



SIGNAL INTREGRITY BETWEEN INTEGRATED DEVICES USING BRIDGE COUPLED TO TOP PORTIONS OF INTEGRATED DEVICES

FIG. 8



SIGNAL INTREGRITY BETWEEN INTEGRATED DEVICES USING NON-BRIDGE CONNECTION

FIG. 7

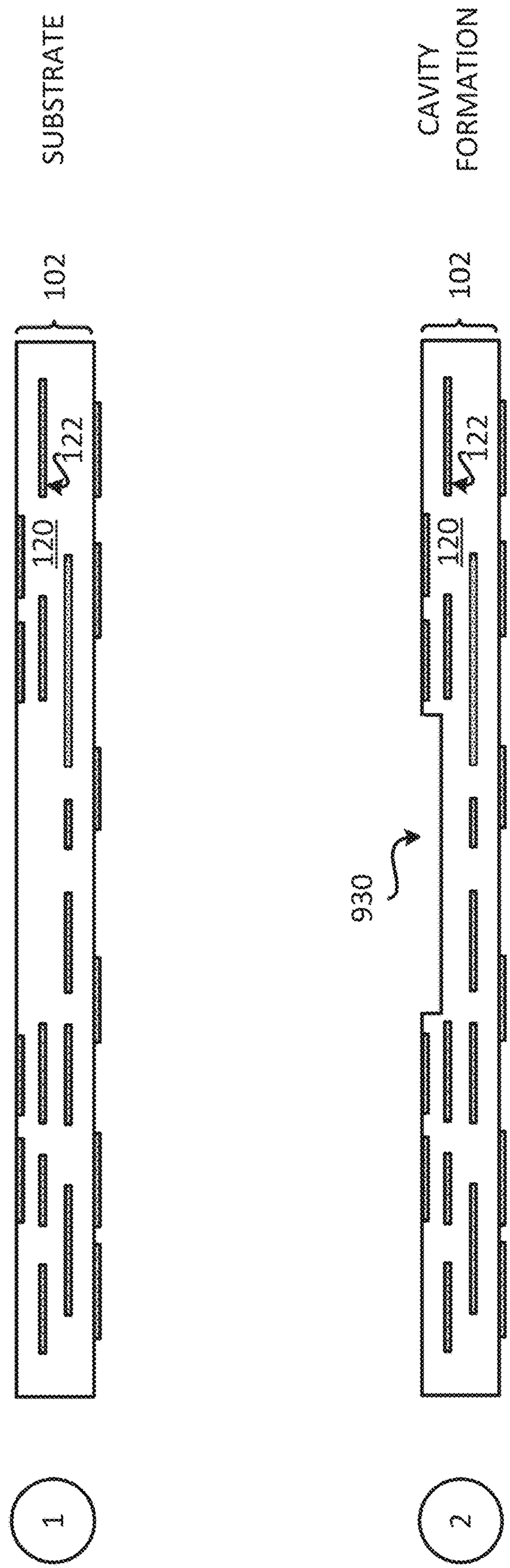


FIG. 9A

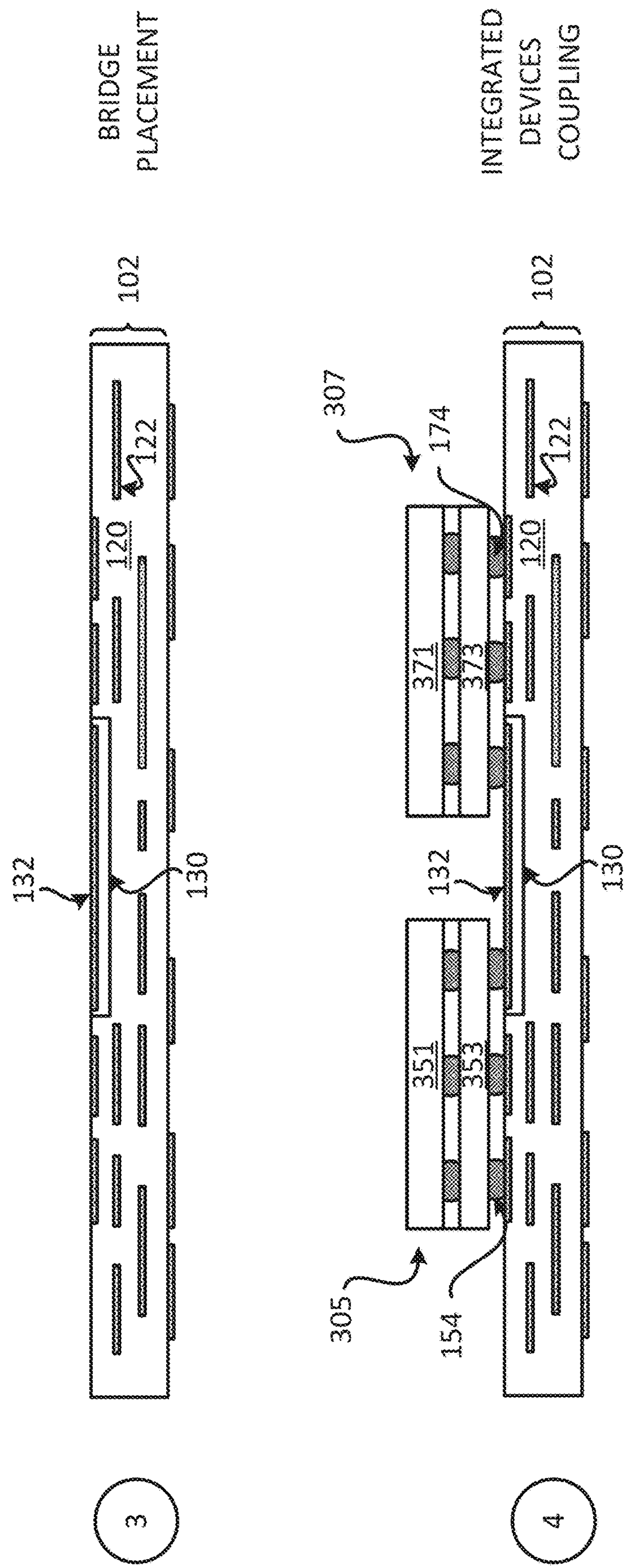

BRIDGE PLACEMENT
INTEGRATED DEVICES COUPLING
102
122
120
130
132
307
174
371
373
351
353
305
154
3
4

FIG. 9B

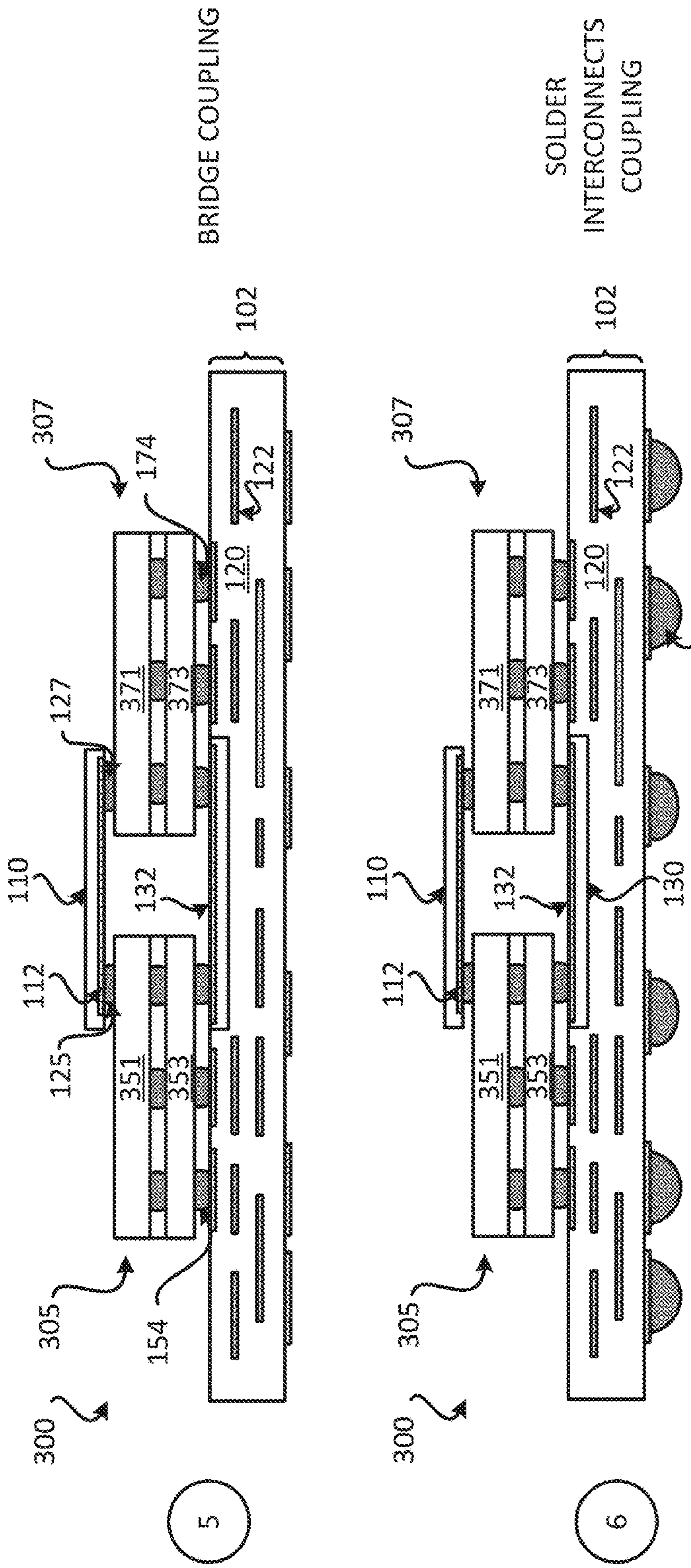

BRIDGE COUPLING
SOLDER INTERCONNECTS COUPLING
102
307
174
122
120
127
371
373
110
132
112
125
351
353
305
154
300
5
124
130
6

FIG. 9C

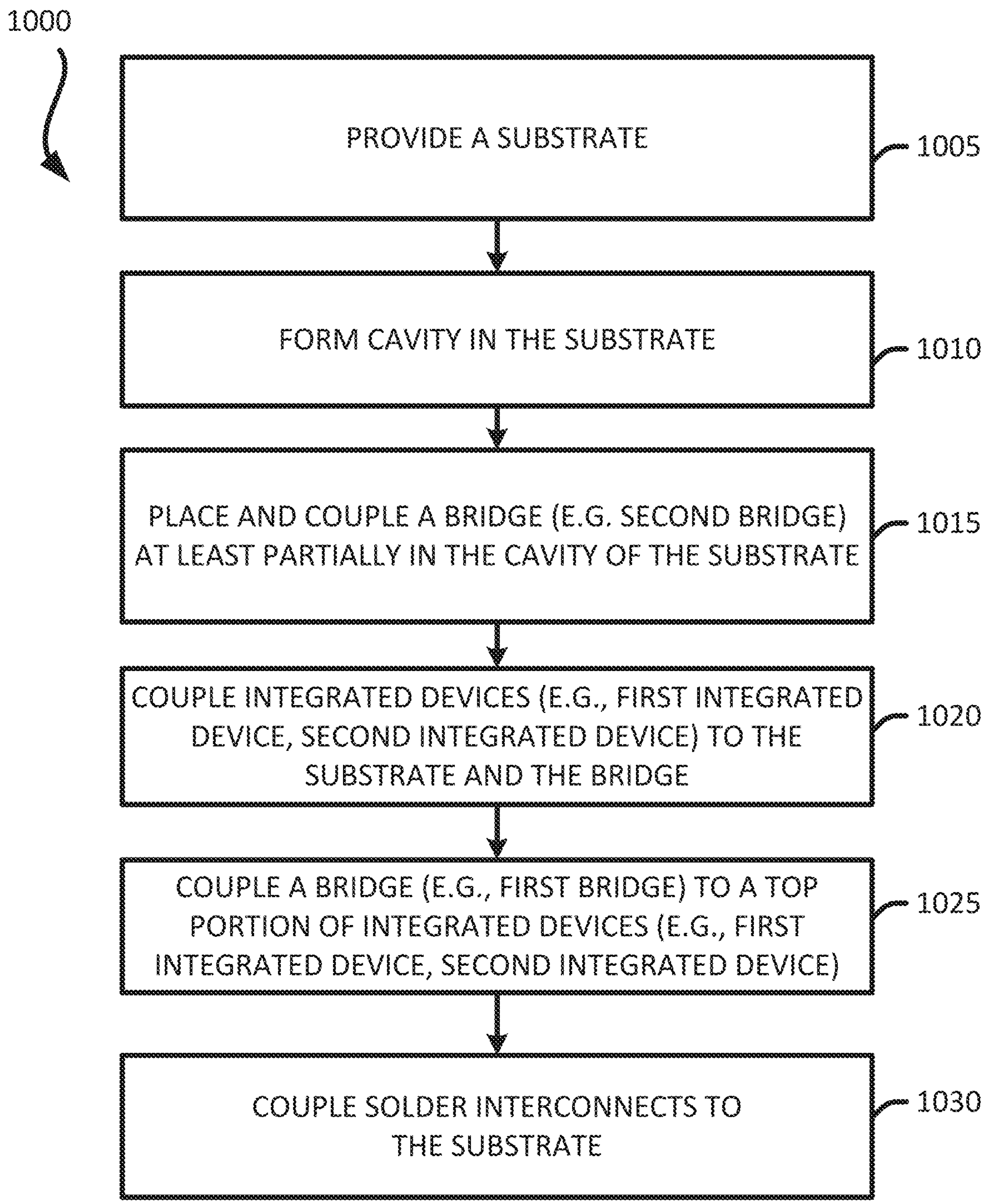

1000
PROVIDE A SUBSTRATE
1005
FORM CAVITY IN THE SUBSTRATE
1010
PLACE AND COUPLE A BRIDGE (E.G. SECOND BRIDGE) AT LEAST PARTIALLY IN THE CAVITY OF THE SUBSTRATE
1015
COUPLE INTEGRATED DEVICES (E.G., FIRST INTEGRATED DEVICE, SECOND INTEGRATED DEVICE) TO THE SUBSTRATE AND THE BRIDGE
1020
COUPLE A BRIDGE (E.G., FIRST BRIDGE) TO A TOP PORTION OF INTEGRATED DEVICES (E.G., FIRST INTEGRATED DEVICE, SECOND INTEGRATED DEVICE)
1025
COUPLE SOLDER INTERCONNECTS TO THE SUBSTRATE
1030

FIG. 10

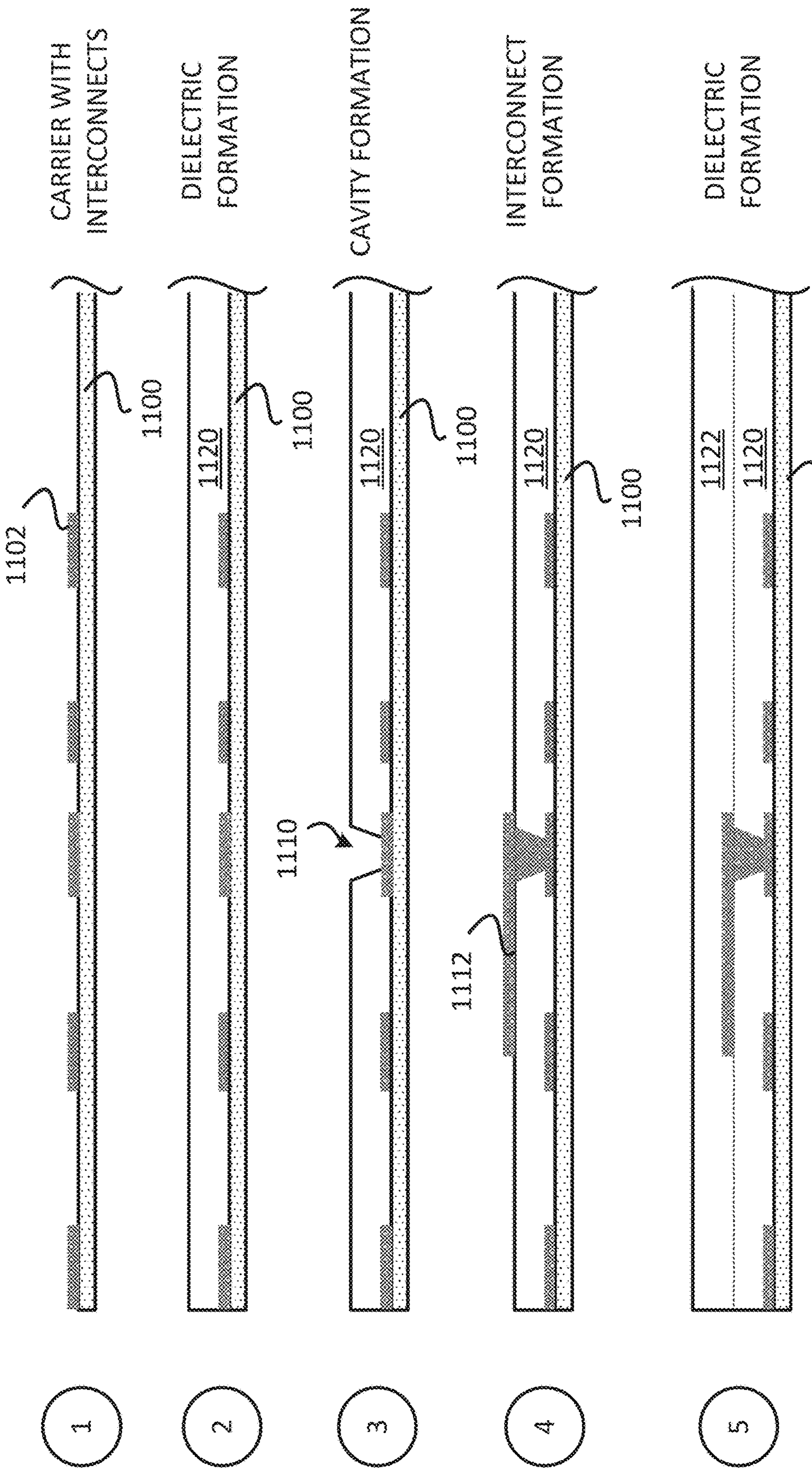
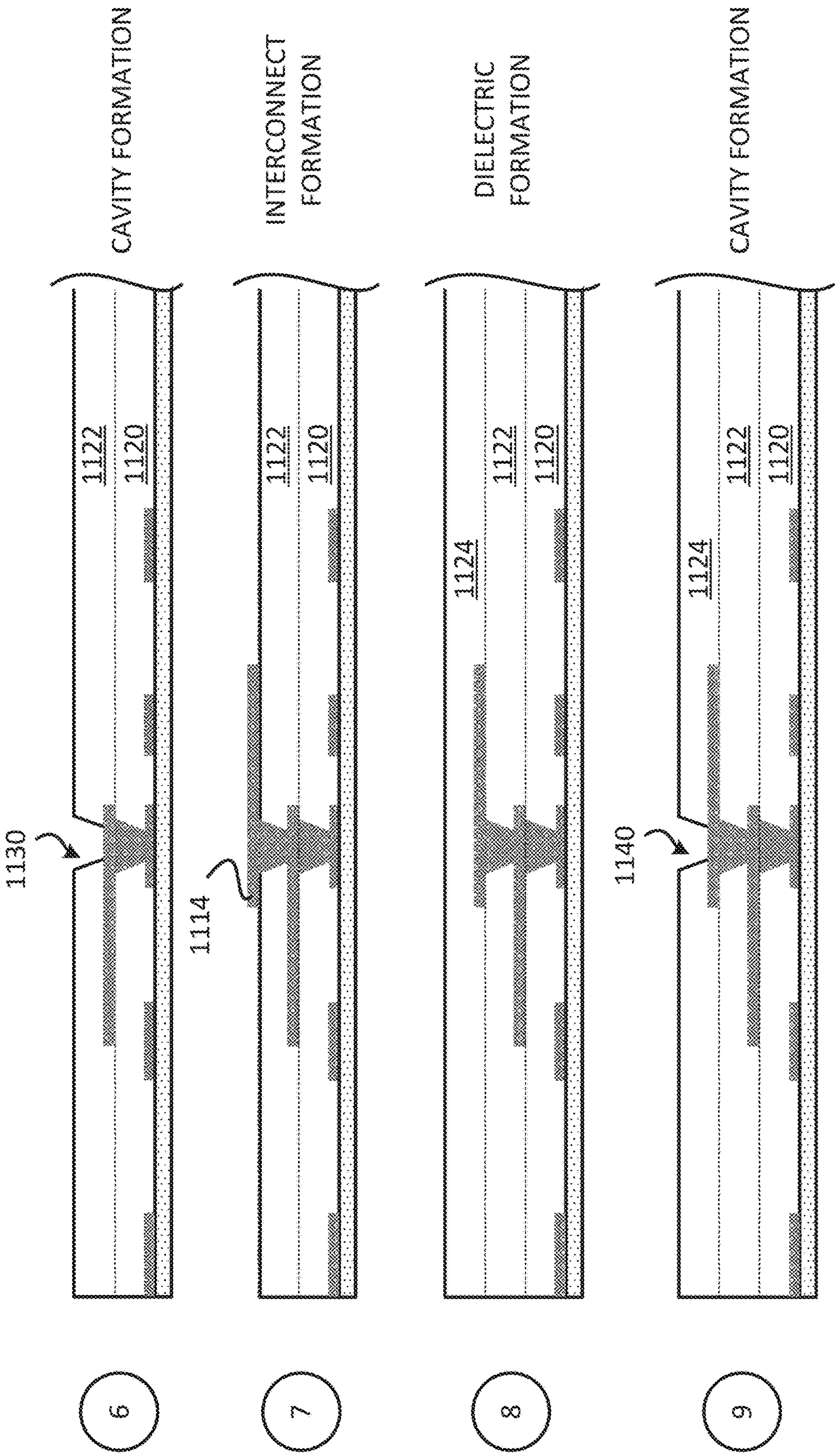

CARRIER WITH INTERCONNECTS
DIELECTRIC FORMATION
CAVITY FORMATION
INTERCONNECT FORMATION
DIELECTRIC FORMATION
1102
1100
1120
1110
1112
1122
1
2
3
4
5

FIG. 11A



CAVITY FORMATION
INTERCONNECT FORMATION
DIELECTRIC FORMATION
CAVITY FORMATION
1122
1120
1122
1120
1124
1122
1120
1124
1122
1120
1130
1114
1140
6
7
8
9

FIG. 11B

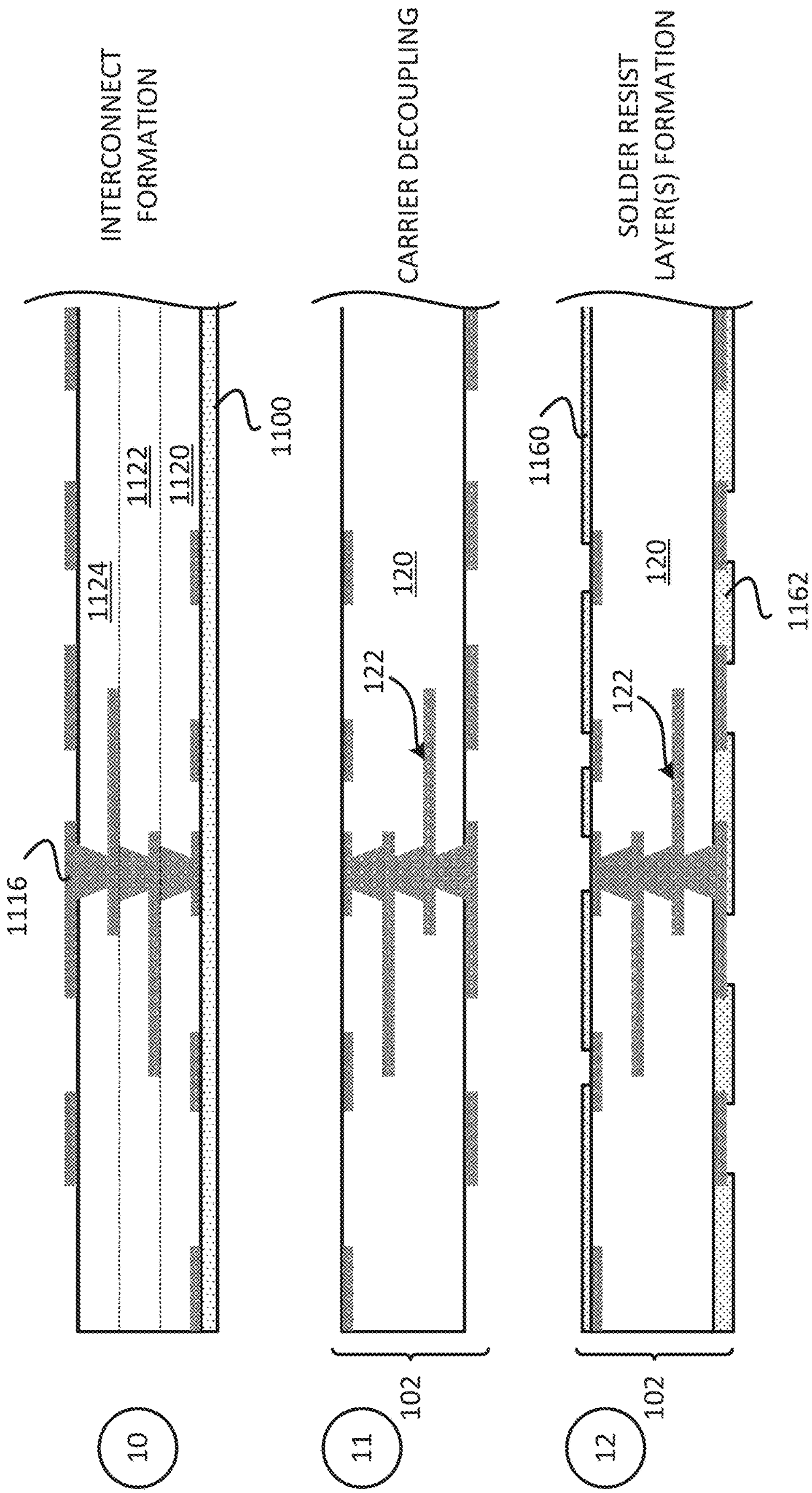

INTERCONNECT FORMATION
CARRIER DECOUPLING
SOLDER RESIST LAYER(S) FORMATION
1100
1122
1120
1124
1116
120
122
1160
1162
102
10
11
12

FIG. 11C

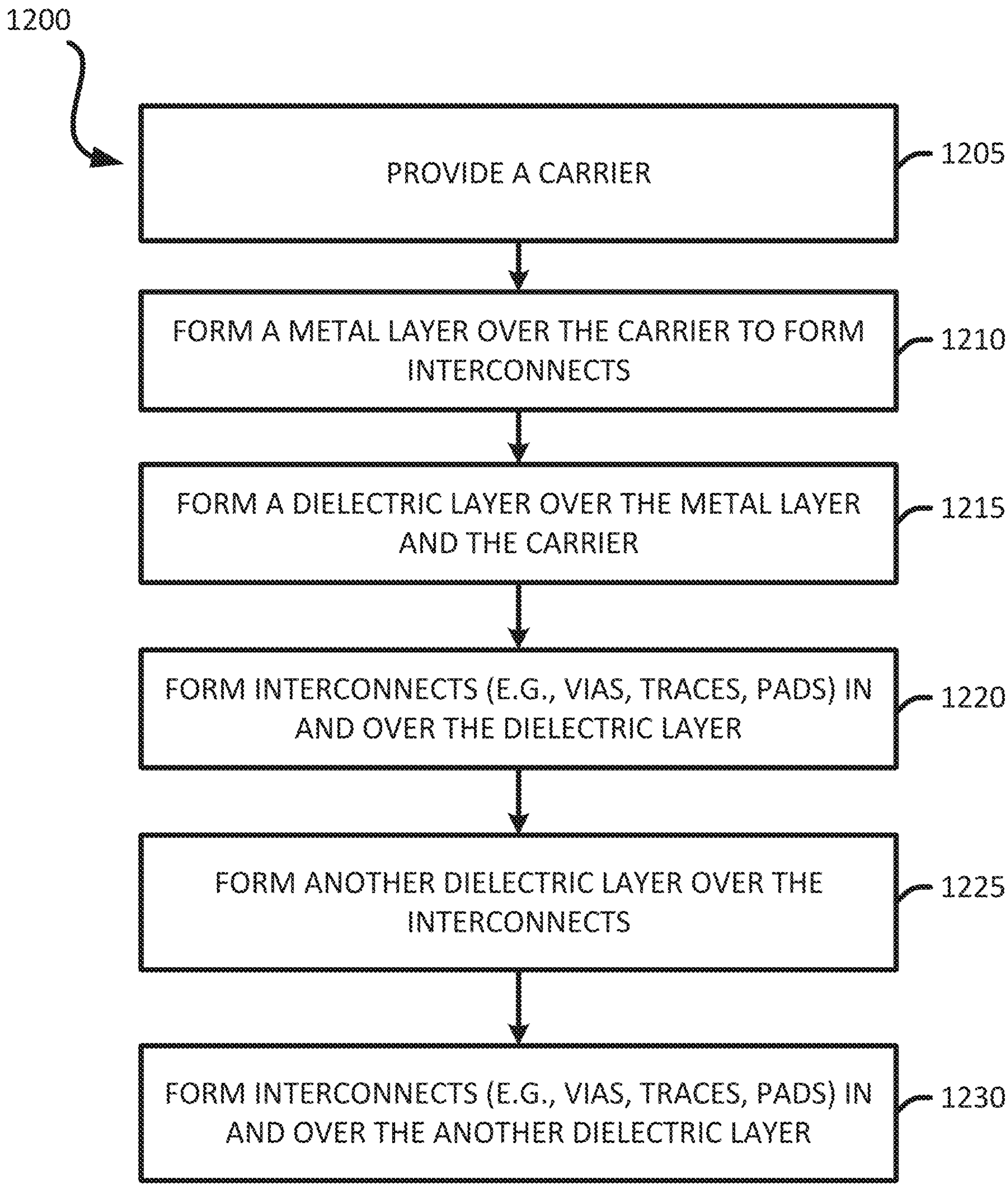

1200
PROVIDE A CARRIER
1205
FORM A METAL LAYER OVER THE CARRIER TO FORM INTERCONNECTS
1210
FORM A DIELECTRIC LAYER OVER THE METAL LAYER AND THE CARRIER
1215
FORM INTERCONNECTS (E.G., VIAS, TRACES, PADS) IN AND OVER THE DIELECTRIC LAYER
1220
FORM ANOTHER DIELECTRIC LAYER OVER THE INTERCONNECTS
1225
FORM INTERCONNECTS (E.G., VIAS, TRACES, PADS) IN AND OVER THE ANOTHER DIELECTRIC LAYER
1230

FIG. 12

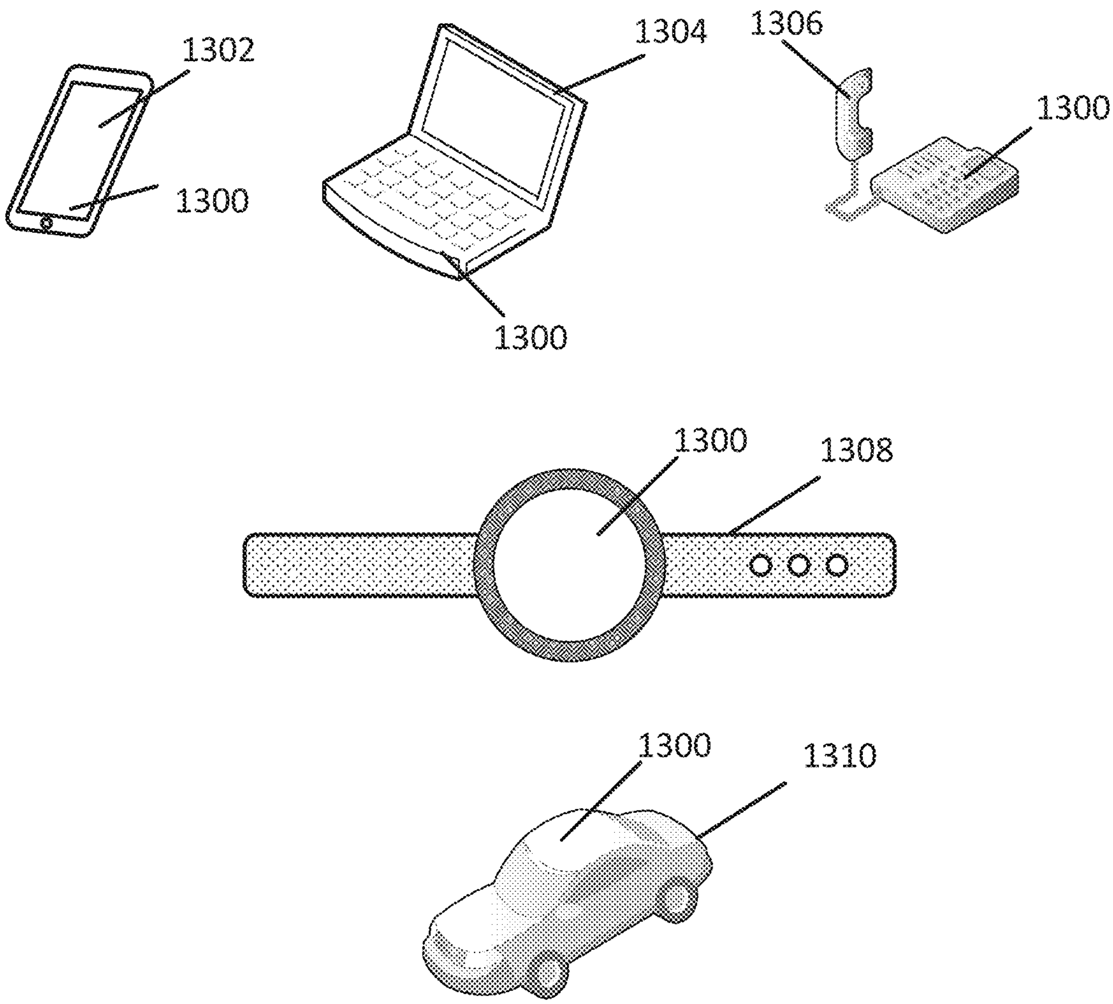


FIG. 13

PACKAGE COMPRISING INTEGRATED DEVICES AND BRIDGE COUPLING TOP SIDES OF INTEGRATED DEVICES

CROSS-REFERENCE TO RELATED APPLICATION

The present application claims priority to and is a continuation application to U.S. patent application Ser. No. 17/357,811, filed on Jun. 24, 2021 in the U.S. Patent and Trademark Office, the entire content of U.S. patent application Ser. No. 17/357,811 is incorporated herein by reference as if fully set forth below and for all applicable purposes.

FIELD

Various features relate to packages comprising integrated devices

BACKGROUND

A package may include a substrate and several integrated devices. The integrated devices may be in communication with each other through the substrate. That is, electrical input and output signals may travel between integrated devices through the substrate. The performance of the package may be tied to how fast these electrical signals may travel between integrated devices and/or the integrity of the signals between the integrated devices. There is an ongoing need to provide better performing packages.

SUMMARY

Various features relate to packages comprising integrated devices.

One example provides a package comprising a substrate; a first integrated device coupled to the substrate through at least a first plurality of solder interconnects; a second integrated device coupled to the substrate through at least a second plurality of solder interconnects; a first bridge coupled to the first integrated device and the second integrated device through at least a third plurality of solder interconnects, wherein the first bridge is configured to provide at least one first electrical path between the first integrated device and the second integrated device, and wherein the first bridge is coupled to a top portion of the first integrated device and a top portion of the second integrated device, through at least the third plurality of solder interconnects; and a second bridge coupled to the first integrated device and the second integrated device through a fourth plurality of solder interconnects, wherein the second bridge is configured to provide at least one second electrical path between the first integrated device and the second integrated device.

Another example provides a package comprising a substrate; a first stack of dies comprising: a first die; and a second die coupled to the first die through at least a first plurality of solder interconnects, wherein the second die is coupled to the substrate through at least a second plurality of solder interconnects; a second stack of dies comprising: a third die; and a fourth die coupled to the third die through at least a third plurality of solder interconnects, wherein the fourth die is coupled to the substrate through at least a fourth plurality of solder interconnects; a first bridge coupled to the first stack of dies and the second stack of dies through at least a fifth plurality of solder interconnects, wherein the first bridge is configured to provide at least one first electrical path between the first stack of dies and the second stack of dies, and wherein the first bridge is coupled to a top portion of the first stack of dies and a top portion of the second stack of dies, through at least the fifth plurality of solder interconnects; and a second bridge coupled to the first stack of dies and the second stack of dies through a sixth plurality of solder interconnects, wherein the second bridge is configured to provide at least one second electrical path between the first stack of dies and the second stack of dies.

Another example provides a method for fabricating a package. The method provides a substrate. The method couples a first integrated device to the substrate. The method couples a second integrated device to the substrate. The method couples a first bridge to the first integrated device and the second integrated device. The first bridge is configured to provide at least one first electrical path between the first integrated device and the second integrated device. The first bridge is coupled to a top portion of the first integrated device and a top portion of the second integrated device. The method couples a second bridge to the first integrated device and the second integrated device. The second bridge is configured to provide at least one second electrical path between the first integrated device and the second integrated device.

BRIEF DESCRIPTION OF THE DRAWINGS

Various features, nature and advantages may become apparent from the detailed description set forth below when taken in conjunction with the drawings in which like reference characters identify correspondingly throughout.

FIG. **1** illustrates a cross sectional profile view of a package that includes a bridge coupled to integrated devices.

FIG. **2** illustrates a cross sectional profile view of another package that includes a bridge coupled to integrated devices.

FIG. **3** illustrates a cross sectional profile view of another package that includes a bridge coupled to integrated devices.

FIG. **4** illustrates a cross sectional profile view of another package that includes a bridge coupled to stacked dies.

FIG. **5** illustrates a cross sectional profile view of a bridge die.

FIG. **6** illustrates a cross sectional profile view of a bridge structure.

FIG. **7** illustrates an exemplary graph of signal integrity for signals between integrated devices using a non-bridge connection.

FIG. **8** illustrates an exemplary graph of signal integrity for signals between integrated devices using a bridge connection.

FIGS. **9**A-**9**C illustrate an exemplary sequence for fabricating a package that includes a bridge coupled to integrated devices.

FIG. **10** illustrates an exemplary flow diagram of a method for fabricating a package that includes a bridge coupled to integrated devices.

FIGS. **11**A-**11**C illustrate an exemplary sequence for fabricating a substrate.

FIG. **12** illustrates an exemplary flow diagram of a method for fabricating a substrate.

FIG. **13** illustrates various electronic devices that may integrate a die, an electronic circuit, an integrated device, an integrated passive device (IPD), a passive component, a package, and/or a device package described herein.

DETAILED DESCRIPTION

In the following description, specific details are given to provide a thorough understanding of the various aspects of the disclosure. However, it will be understood by one of ordinary skill in the art that the aspects may be practiced without these specific details. For example, circuits may be shown in block diagrams in order to avoid obscuring the aspects in unnecessary detail. In other instances, well-known circuits, structures and techniques may not be shown in detail in order not to obscure the aspects of the disclosure.

The present disclosure describes a package comprising a substrate, a first integrated device coupled to the substrate, a second integrated device coupled to the substrate, a first bridge and a second bridge. The first bridge is coupled to the first integrated device and the second integrated device. The first bridge is configured to provide at least one first electrical path between the first integrated device and the second integrated device. The first bridge is coupled to a top portion of the first integrated device and a top portion of the second integrated device. The second bridge is coupled to the first integrated device and the second integrated device. The second bridge is configured to provide at least one second electrical path between the first integrated device and the second integrated device. The second bridge may be located at least partially in the substrate. The second bridge may be located above the substrate. The first integrated device may include one or more dies. The second integrated device may include one or more dies. The package may provide better signal integrity for input/output signals between integrated devices, resulting in a better performing package.

Exemplary Packages Comprising a Bridge Coupled to Integrated Devices

FIG. **1** illustrates a cross sectional profile view of a package **100** that includes a substrate **102**, an integrated device **105** (e.g., first integrated device), an integrated device **107** (e.g., second integrated device), a bridge **110**, and a bridge **130**. The package **100** is coupled to a board **106** through a plurality of solder interconnects **124**. The board **106** may include a printed circuit board (PCB).

The substrate **102** includes at least one dielectric layer **120** and a plurality of interconnects **122**. Different substrates may have different numbers of metal layers. Different implementations may use different substrates for the substrate **102**. The substrate **102** may include an embedded trace substrate (ETS), a laminate substrate, a coreless substrate and/or a cored substrate. The substrate **202** may be fabricated using different processes, including an ETS process, a semi-additive process (SAP), and/or a modified semi-additive process (mSAP).

The bridge **130** may be located at least partially in the substrate **102**. In some implementations, the bridge **130** may be located in a cavity of the substrate **102**. The bridge **130** may be embedded in the substrate **102**. The bridge **130** may be a second bridge. The bridge **130** may include at least one bridge interconnect **132**. As will be further described below, the bridge **130** may include a bridge die and/or bridge structure. The bridge **130** may be a means for second bridge interconnection. In some implementations, the bridge **130** may be located above the substrate **102**. In some implementations, the bridge **130** is not located in the substrate **102**.

The integrated device **105** is coupled to the substrate **102** through a plurality of solder interconnects **154**. The integrated device **105** is also coupled to the bridge **130** through at least one solder interconnect **154*a*** from the plurality of solder interconnect **154**. The integrated device **107** is coupled to the substrate **102** through a plurality of solder interconnects **174**. The integrated device **107** is also coupled to the bridge **130** through at least one solder interconnect **174*a*** from the plurality of solder interconnect **174**. The bridge **130** may be configured to provide at least one second electrical path between the integrated device **105** and the integrated device **107**. For example, the at least one second electrical path between the integrated device **105** and the integrated device **107** may include the at least one solder interconnect **154*a***, the at least one bridge interconnect **132** and/or the at least one solder interconnect **174*a***.

The bridge **110** may be coupled to a top portion of the integrated device **105** through at least one solder interconnect **125**. The bridge **110** may also be coupled to a top portion of the integrated device **107** through at least one solder interconnect **127**. The bridge **110** may be a first bridge. The bridge **110** may include at least one bridge interconnect **112**. As will be further described below, the bridge **110** may include a bridge die and/or bridge structure. The bridge **110** may be a means for first bridge interconnection. The bridge **110** may be configured to provide at least one first electrical path between the integrated device **105** and the integrated device **107**. For example, the at least one first electrical path between the integrated device **105** and the integrated device **107** may include the at least one solder interconnect **125**, the at least one bridge interconnect **112** and/or the at least one solder interconnect **127**. In some implementations, one or more input/output (I/O) signals between the integrated device **105** and the integrated device **107** may be configured to travel through the bridge **110** (e.g., configured to travel through the at least one solder interconnect **125**, the at least one bridge interconnect **112** and the at least one solder interconnect **127**). In some implementations, input/output signals that travel through the bridge **110** may have better signal integrity because the paths are located farther away from interconnects configured as electrical paths for other current (e.g., power). In some implementations, the bridge **110** provides a short electrical path between the integrated device **105** and the integrated device **107**, thus providing improved package performance. In some implementations, the bridge **110** frees up solder interconnects to the substrate to be used for power and/or ground, resulting in improved power distribution network (PDN) performance. In some implementations, signals between the integrated device **105** and the integrated device **107** may be configured to travel through a back side of the integrated device **105** and/or the back side of the integrated device **107**. In some implementations, the top portion of the integrated device **105** may include the back side of the integrated device **105**. In some implementations, the top portion of the integrated device **107** may include the back side of the integrated device **107**. The back side of an integrated device may be the side of the integrated device that includes a substrate (e.g., silicon). The front side of an integrated device may be a side opposite to the back side of an integrated device.

FIG. **2** illustrates a cross sectional profile view of a package **200** that includes the substrate **102**, the integrated device **105** (e.g., first integrated device), the integrated device **107** (e.g., second integrated device), the bridge **110**, and the bridge **130**. The package **200** is similar to the package **100**. However, the bridge **110** and the bridge **130** may be coupled to the integrated device **105** and the integrated device **107** through a plurality of pillar interconnects.

The integrated device **105** is coupled to the substrate **102** through a plurality of pillar interconnects **254** and the plurality of solder interconnects **154**. The integrated device **105** is coupled to the bridge **130** through at least one pillar interconnect **254*a*** and at least one solder interconnect **154*a***. The integrated device **107** is coupled to the substrate **102** through a plurality of pillar interconnects **274** and the plurality of solder interconnects **174**. The integrated device

107 is coupled to the bridge **130** through at least one pillar interconnect **274***a* and at least one solder interconnect **174***a*. At least one second electrical path between the integrated device **105** and the integrated device **107** may include the at least one pillar interconnect **254***a*, the at least one solder interconnect **154***a*, the at least one bridge interconnect **132** (of the bridge **130**), the at least one pillar interconnect **274***a*, and/or the at least one solder interconnect **174***a*.

The bridge **110** is coupled to a top portion of the integrated device **105** through at least one pillar interconnect **225** and at least one solder interconnect **125**. The bridge **110** is coupled to a top portion of the integrated device **107** through at least one pillar interconnect **227** and at least one solder interconnect **127**. At least one first electrical path between the integrated device **105** and the integrated device **107** may include the at least the at least one solder interconnect **125**, one pillar interconnect **225**, the at least one bridge interconnect **112** (of the bridge **110**), the at least one pillar interconnect **227**, and/or the at least one solder interconnect **127**.

As will be further described below, the integrated device **105** may include one or more integrated devices (e.g., dies). Similarly, the integrated device **107** may include one or more integrated devices (e.g., dies).

FIG. **3** illustrates a cross sectional profile view of a package **300** that includes the substrate **102**, an integrated device **305** (e.g., first integrated device), an integrated device **307** (e.g., second integrated device), the bridge **110**, and the bridge **130**.

The integrated device **305** may be a package that includes a first die **351**, a second die **353** and an encapsulation layer **356**. The first die **351** and/or the second die **353** may be examples of integrated devices. The second die **353** may be coupled to the first die **351** through a plurality of solder interconnects **354**. The first die **351** may be located over (e.g., above) the second die **353**. The encapsulation layer **356** may be located between the first die **351** and the second die **353**. In some implementations, the encapsulation layer **356** may at least partially encapsulate the first die **351** and/or the second die **353**. The integrated device **305** is coupled to the substrate **102** and the bridge **130** through a plurality of solder interconnects **154**.

As will be further described below, the first die **351** and the second die **353** may be aligned and positioned in different directions in the integrated device **305**. The first die **351** may include a front side and a back side. The second die **353** may also include a front side and a back side. In some implementations, the front side of the first die **351** may face the front side of the second die **353**. In some implementations, the front side of the second die **353** may face the back side of the first die **351**. In some implementations, the back side of the second die **353** may face the front side of the first die **351**. A back side of a die may be a side of the die that includes a die substrate (e.g., silicon). The front side of a die may be a side opposite to the back side of a die.

The integrated device **307** may be a package that includes a third die **371**, a fourth die **373** and an encapsulation layer **376**. The third die **371** and/or the fourth die **373** may be examples of integrated devices. The fourth die **373** may be coupled to the third die **371** through a plurality of solder interconnects **374**. The third die **371** may be located over (e.g., above) the fourth die **373**. The encapsulation layer **376** may be located between the third die **371** and the fourth die **373**. In some implementations, the encapsulation layer **376** may at least partially encapsulate the third die **371** and/or the fourth die **373**. The integrated device **307** is coupled to the substrate **102** and the bridge **130** through a plurality of solder interconnects **174**. An encapsulation layer (e.g., **356**, **376**) may include a mold, a resin and/or an epoxy. An encapsulation layer (e.g., **356**, **376**) may be a means for encapsulation.

As will be further described below, the third die **371** and the fourth die **373** may be aligned and positioned in different directions in the integrated device **307**. The third die **371** may include a front side and a back side. The fourth die **373** may also include a front side and a back side. In some implementations, the front side of the third die **371** may face the front side of the fourth die **373**. In some implementations, the front side of the fourth die **373** may face the back side of the third die **371**. In some implementations, the back side of the fourth die **373** may face the front side of the third die **371**.

Different implementations may position the dies in the package **300** differently. In some implementations, the back side of the first die **351** and/or the back side of the third die **371** may face the bridge **110**. In some implementations, the front side of the first die **351** and/or the front side of the third die **371** may face the bridge **110**. In some implementations, the back side of the second die **353** and/or the back side of the fourth die **373** may face the bridge **130** and/or the substrate **102**. In some implementations, the front side of the second die **353** and/or the front side of the fourth die **373** may face the bridge **130** and/or the substrate **102**.

The bridge **110** is coupled to the first die **351** through the at least one solder interconnect **125**. The bridge **110** is coupled to the third die **371** through the at least one solder interconnect **127**. The first die **351** may be configured to be electrically coupled to the third die **371** through the bridge **110**. At least one first electrical path (e.g., for input/output signals) between the first die **351** and the third die **371** may include the at least one solder interconnect **125**, the at least one bridge interconnect **112**, and/or the at least one solder interconnect **127**. The bridge **110** provide a shorter electrical path between the first die **351** and the third die **371**, by not having to travel through the second die **353**, the substrate **102**, the bridge **130** and/or the fourth die **373**.

The bridge **130** is coupled to the second die **353** through the at least one solder interconnect **154***a*. The bridge **130** is coupled to the fourth die **373** through the at least one solder interconnect **174***a*. The second die **353** may be configured to be electrically coupled to the fourth die **373** through the bridge **130**. At least one second electrical path (e.g., for input/output signals) between the second die **353** and the fourth die **373** may include the at least one solder interconnect **154***a*, the at least one bridge interconnect **132**, and/or the at least one solder interconnect **174***a*.

In some implementations, power to the integrated device **305**, including power to the first die **351** and/or the second die **353** may be provided through the substrate **102** (e.g., plurality of interconnects **122**) and at least one solder interconnect **154**. In some implementations, power to the first die **351** may be provided through the substrate **102** (e.g., plurality of interconnects **122**), at least one solder interconnect **154**, the second die **353** and at least one solder interconnect **354**. In some implementations, power to the integrated device **307**, including power to the third die **371** and/or the fourth die **373** may be provided through the substrate **102** (e.g., plurality of interconnects **122**) and at least one solder interconnect **174**. In some implementations, power to the third die **371** may be provided through the substrate **102** (e.g., plurality of interconnects **122**), at least one solder interconnect **174**, the fourth die **373** and at least one solder interconnect **374**. It is noted that the integrated device **305** may include more than two stacked dies. Similarly, the integrated device **307** may include more than two stacked dies.

FIG. **4** illustrates a cross sectional profile view of an implementation of the package **300**. As shown in FIG. **4**, the first die **351** is coupled to the second die **353** such that the front side of the first die **351** faces the front side of the second die **353**. The back side of a die may be a side that includes a die substrate (e.g., silicon). The front side of a die may be a side that is opposite to the back side of a die. The first die **351** includes a plurality of through substrate vias (TSVs) **451**. The plurality of TSVs **451** may be located in the back side of the first die **351**. The second die **353** includes a plurality of through substrate vias (TSVs) **453**. The plurality of TSVs **453** may be located in the back side of the second die **353**.

Similarly, the third die **371** is coupled to the fourth die **373** such that the front side of the third die **371** faces the front side of the fourth die **373**. The third die **371** includes a plurality of through substrate vias (TSVs) **471**. The plurality of TSVs **471** may be located in the back side of the third die **371**. The fourth die **373** includes a plurality of through substrate vias (TSVs) **473**. The plurality of TSVs **473** may be located in the back side of the fourth die **373**.

In some implementations, the back side of the first die **351** and the back side of the third die **371** may face the bridge **110**. In some implementations, the front side of the first die **351** and the front side of the third die **371** may face the substrate **102** and/or the bridge **130**. In some implementations, at least one electrical current (e.g., input/output signal) may travel between the first die **351** and the third die **371** through the plurality of TSVs **451**, the at least one solder interconnect **125**, the at least one bridge interconnect **112**, the at least one solder interconnect **127**, and the plurality of TSVs **471**.

In some implementations, at least one electrical current (e.g., input/output signal) may travel between the second die **353** and the fourth die **373** through the plurality of TSVs **453**, the at least one solder interconnect **154*a***, the at least one bridge interconnect **132**, the at least one solder interconnect **174*a***, and the plurality of TSVs **473**.

In some implementations, the first die **351** and the second die **353** may each be configured to be electrically coupled to the substrate **102**. In some implementations, at least one electrical current (e.g., power) between the first die **351** and the substrate **102** may travel through the plurality of solder interconnect **154**, the plurality of TSVs **453**, the second die **353** and the plurality of solder interconnects **354**. In some implementations, at least one electrical current (e.g., power) between the second die **353** and the substrate **102** may travel through the plurality of solder interconnect **154** and the plurality of TSVs **453**.

In some implementations, the third die **371** and the fourth die **373** may each be configured to be electrically coupled to the substrate **102**. In some implementations, at least one electrical current (e.g., power) between the third die **371** and the substrate **102** may travel through the plurality of solder interconnect **174**, the plurality of TSVs **473**, the fourth die **373** and the plurality of solder interconnects **354**. In some implementations, at least one electrical current (e.g., power) between the fourth die **373** and the substrate **102** may travel through the plurality of solder interconnect **174** and the plurality of TSVs **473**.

FIG. **4** illustrates one example of how stacked dies may be implemented. However, as mentioned above, the dies may be arranged in different directions and may include different numbers of dies. The disclosure illustrates and describes the bridge **130** as being located at least partially (e.g., embedded) in the substrate **102**. However, in some implementations, the bridge **130** may be located over the substrate **102**. For example, the bridge **130** may be located between integrated devices and the substrate **102**. Thus, in some implementations, the bridge **130** is not located in the substrate **102**.

FIGS. **3** and **4** illustrate examples of packages where the first integrated device (e.g., **305**) includes two dies and the second integrated device (**307**) includes two dies. However, a package may include other configurations and combinations of integrated devices. For example, a first integrated device may include one die and a second integrated device may include two dies. In another example, a first integrated device may include two dies and a second integrated device may include three or more dies. In some implementations, a package may include a first integrated device, a second integrated device, a third integrated device and a fourth integrated device, that are all coupled to a substrate. A bridge may be coupled to the top portions of the first integrated device and the second integrated device. Another bridge may be coupled to the top portions of the third integrated device and the fourth integrated device. A top portion of an integrated device may include a side of the integrated device that faces away from a substrate.

An integrated device (e.g., **105**, **107**, **305**, **307**) may include a die (e.g., semiconductor bare die). The integrated device may include a chip, a chiplet, a radio frequency (RF) device, a passive device, a filter, a capacitor, an inductor, an antenna, a transmitter, a receiver, a GaAs based integrated device, a surface acoustic wave (SAW) filters, a bulk acoustic wave (BAW) filter, a light emitting diode (LED) integrated device, a silicon (Si) based integrated device, a silicon carbide (SiC) based integrated device, a processor, a memory, a power management integrated device and/or combinations thereof. An integrated device (e.g., **105**, **107**, **305**, **307**) may include at least one electronic circuit (e.g., first electronic circuit, second electronic circuit, etc. . . .).

FIG. **5** illustrates a view of a bridge **500**. The bridge **500** may be a bridge die. The bridge **500** may be a passive bridge die. The bridge **500** may represent the bridge **110** and/or the bridge **130** in the disclosure. The bridge **500** may be a means for bridge interconnection. The bridge **500** includes a die substrate **510** (e.g., bridge die substrate), at least one bridge interconnect **512**, a passivation layer **520**, a plurality of under bump interconnects **514** and a plurality of solder interconnect **530**. The die substrate **510** may include silicon (Si). The at least one bridge interconnect **512** may be coupled to the plurality of under bump interconnects **514**. In some implementations, the at least one bridge interconnect **512** may include rows of bridge interconnects. The plurality of solder interconnects **530** may be coupled to the plurality of under bump interconnects **514**. It is noted that plurality of under bump interconnects **514** may considered part of the bridge interconnect for the bridge **500**. Thus, a bridge interconnect may also refer to at least one under bump interconnect of a bridge. The bridge **500** is configured to allow one or more electrical currents (e.g., input/output signals) to travel through the under bump interconnect **514*a***, the at least one bridge interconnect **512** and the under bump interconnect **514*b***.

FIG. **6** illustrates a view of a bridge **600**. The bridge **600** may be a bridge structure and/or a bridge substrate. The bridge **600** may represent the bridge **110** and/or the bridge **130** in the disclosure. The bridge **600** may be a means for bridge interconnection. The bridge **600** includes at least one dielectric layer **610**, at least one bridge interconnect **612** and a plurality of solder interconnect **530**. The at least one bridge interconnect **612** includes a bridge interconnect **612***a* (e.g., bridge via, bridge pad), a bridge interconnect **612***c* (e.g., bridge trace) and a bridge interconnect **612***b* (e.g., bridge via, bridge pad). The bridge **600** is configured to allow one or more electrical currents (e.g., input/output signals) to travel through the bridge interconnect **612***a*, the bridge interconnect **612***c* and the bridge interconnect **612***b*. In some implementations, there be several rows of the bridge interconnect **612***a*, the bridge interconnect **612***c* and the bridge interconnect **612***b*.

FIGS. **7** and **8** illustrate graphs of exemplary signal integrity between two integrated devices. FIG. **7** illustrates an exemplary graph **700** of signal integrity between two integrated devices, where the signals travel through a non-bridge connection. FIG. **8** illustrates an exemplary graph **800** of signal integrity between two integrated devices, where the signals travel through a bridge coupled to top portions of integrated devices. The graph **700** illustrates an eye opening **710**. The graph **800** illustrates an eye opening **810**. The eye opening **810** is greater than the eye opening **710**, which may indicate that signals traveling through the bridge may have better signal integrity, including signals with less cross talk, improved differential insertion loss, improved intra-pair skew and/or improved mode conversion, relative to signals that may not be traveling through a bridge. These improvements in signal integrity may be because of shorter routing distances between integrated devices, and/or more availability of power and ground bumps for the integrated devices. That is, bumps and/or solder interconnects between the integrated devices and a substrate that would normally be used for signals routing may be used instead for power and ground, which may ultimately help improve the power distribution network (PDN) performance of the integrated devices and the package.

Having described various packages, a sequence for fabricating a package will now be described below.

Exemplary Sequence for Fabricating a Packages Comprising a Bridge Coupled to Integrated Devices

FIGS. **9**A-**9**C illustrate an exemplary sequence for providing or fabricating a package that includes a bridge coupled to integrated devices. In some implementations, the sequence of FIGS. **9**A-**9**C may be used to provide or fabricate the package **300** of FIG. **3**, or any of the packages described in the disclosure.

It should be noted that the sequence of FIGS. **9**A-**9**C may combine one or more stages in order to simplify and/or clarify the sequence for providing or fabricating a package. In some implementations, the order of the processes may be changed or modified. In some implementations, one or more of processes may be replaced or substituted without departing from the spirit of the disclosure. Different implementations may fabricate a package differently.

Stage 1, as shown in FIG. **9**A, illustrates a state after a substrate **102** is provided. The substrate **102** may be fabricated or provided by a supplier. The substrate **102** includes at least one dielectric layer **120** and a plurality of interconnects **122**. The substrate **102** may include different numbers of metal layers. The substrate **102** may include a laminated substrate, a cored substrate and/or a coreless substrate (e.g., ETS). An example of fabricating a substrate is further described below in at least FIGS. **11**A-**11**C.

Stage 2 illustrates a state after a cavity **930** is formed in the substrate **102**. Different implementations may form the cavity **930** differently. In some implementations, a laser process (e.g., laser ablation) may be used to form the cavity **930**. Different implementations may have a cavity with different sizes, shapes and depths. It is noted that the cavity **930** may already be formed in the substrate **102** when the substrate **102** is provided at Stage 1 of FIG. **9**A.

Stage 3, as shown in FIG. **9**B, illustrates a state after the bridge **130** is placed at least partially in the cavity **930** of the substrate **102**. In some implementations, an adhesive (which is not shown) may be used to place and couple the bridge **130** to the substrate **102**. The bridge **130** includes at least one bridge interconnect **132**. The bridge **130** may include the bridge **500** or the bridge **600**. It is noted that bridge **130** may be provided during a different stage of the process. For example, the bridge **130** may be coupled to the integrated devices before the bridge **130** is coupled to the substrate **102**. In some implementations, the bridge **130** may be coupled to integrated devices after the integrated devices have been coupled to the substrate **102**.

Stage 4 illustrates a state a state after the integrated device **305** and the integrated device **307** are coupled to the substrate **102**. The integrated device **305** is coupled to the substrate **102** through the plurality of solder interconnect **154**. The integrated device **305** is also coupled to the bridge **130** through the plurality of solder interconnects **154**. The integrated device **307** is coupled to the substrate **102** through the plurality of solder interconnect **174**. The integrated device **307** is also coupled to the bridge **130** through the plurality of solder interconnects **174**. As shown in Stage 4, the integrated device **305** includes a first die **351** and a second die **353**. Similarly, the integrated device **307** includes a third die **371** and a fourth die **373**. A solder reflow process may be used to couple the integrated device **305** and the integrated device **307** to the substrate **102** and the bridge **130**. It is noted that in some implementations, a plurality of pillar interconnects (as described in FIG. **2**) may be used to couple the integrated device **305** and the integrated device **307** to the substrate **102** and the bridge **130**. Thus, in some implementations, coupling the integrated device **305** and the integrated device **307** to the substrate **102** may also include coupling the integrated device **305** and the integrated device **307** to the bridge **130**. However, in some implementations, the integrated device **305** and the integrated device **307** may be coupled to the bridge **130** during a different sequence. For example, the bridge **130** may be coupled to the integrated device **305** and the integrated device **307**, and then the coupled integrated device **305**, the integrated device **307**, the bridge **130** may be coupled to the substrate **102**. In another example, the bridge **130** may be coupled to the integrated device **305** and the integrated device **307**, after the integrated device **305** and the integrated device **307** have been coupled to the substrate **102**. For example, there may be a cavity in the substrate **102** that extends through the substrate **102**. The integrated device **305** and the integrated device **307** may be coupled to the substrate **102**, and the bridge **130** may be coupled to the integrated device **305** and the integrated device **307** through the cavity that extends through the substrate **102**.

Stage 5, as shown in FIG. **9**C, illustrates a state after the bridge **110** is coupled to a top portion of the integrated device **305** and a top portion of the integrated device **307**. The bridge **110** may be coupled to the integrated device **305** through the at least one solder interconnect **125**. The bridge **110** may be coupled to the integrated device **307** through the at least one solder interconnect **127**. A solder reflow process may be used to couple the bridge **110** to the integrated device **305** and the integrated device **307**. It is noted that in some implementations, a plurality of pillar interconnects (as described in FIG. **2**) may be used to couple the bridge **110** to the integrated device **305** and the integrated device **307**.

Stage 6 illustrates a state the plurality of solder interconnects **124** is coupled to a bottom surface of the substrate **102**. A solder reflow process may be used to couple the plurality of solder interconnects **124** to the substrate **102**. The plurality of solder interconnects **124** may be coupled to pad interconnects from the plurality of interconnects **122**. Stage 6 may illustrate the package **300**.

Exemplary Flow Diagram of a Method for Fabricating a Packages Comprising a Bridge Coupled to Integrated Devices

In some implementations, fabricating a package includes several processes. FIG. **10** illustrates an exemplary flow diagram of a method **1000** for providing or fabricating a package that includes a bridge coupled to integrated devices. In some implementations, the method **1000** of FIG. **10** may be used to provide or fabricate the package **300** of FIG. **3**. However, the method **1000** of FIG. **10** may be used fabricate any package in the disclosure.

It should be noted that the method of FIG. **10** may combine one or more processes in order to simplify and/or clarify the method for providing or fabricating a package. In some implementations, the order of the processes may be changed or modified.

The method provides (at **1005**) a substrate (e.g., **102**). The substrate **102** may include at least one dielectric layer **120** and a plurality of interconnects **122**. The substrate **102** may include different numbers of metal layers. The substrate **102** may include a laminated substrate, a cored substrate and/or a coreless substrate (e.g., ETS). The substrate may be provided by a supplier or fabricated. An example of fabricating a substrate is further described below in at least FIGS. **11**A-**11**C. Stage 1 of FIG. **9**A illustrates and describes an example of providing a substrate.

The method forms (at **1010**) a cavity (e.g., **930**) in the substrate (e.g., **102**). Different implementations may form the cavity **930** differently. In some implementations, a laser process (e.g., laser ablation) may be used to form the cavity **930**. Different implementations may have a cavity with different sizes, shapes and depths. In some implementations, the cavity **930** may extend entirely through the substrate **102**. It is noted that the cavity **930** may already be formed in the substrate **102** when the substrate **102** is provided (at **1005**). Stage 2 of FIG. **9**A illustrates and describes an example of a cavity formed in a substrate.

The method places and couples (at **1015**) a bridge (e.g., **130**) in the cavity **930** of the substrate **102**. The bridge **130** may be placed at least partially in the cavity **930**. In some implementations, an adhesive (which is not shown) may be used to place and couple the bridge **130** to in the cavity **930** of the substrate **102**. The bridge **130** includes at least one bridge interconnect **132**. The bridge **130** may include the bridge **500** or the bridge **600**. It is noted that the bridge **130** may be coupled to the substrate **102** differently. In some implementations, the bridge **130** may be coupled to integrated devices before being coupled to the substrate **102** or before being positioned over the substrate **102**. Stage 3 of FIG. **9**B illustrates and describes an example of a bridge that is placed at least partially in a cavity of a substrate.

The method couples (at **1020**) a plurality of integrated devices (e.g., **105**, **107**, **305**, **307**) to the substrate **102** and the bridge **130**. The plurality of integrated devices may be coupled to the substrate **102** and the bridge **130** through a plurality of solder interconnect (e.g., **154**, **174**) and/or a plurality of pillar interconnects (e.g., **254**, **274**). A solder reflow process may be used to couple the integrated devices (e.g., **105**, **107**, **305**, **307**) to the substrate **102** and the bridge **130**. In some implementations, the bridge **130** may be coupled to the integrated devices before the integrated devices are coupled to the substrate **102**. It is noted that in some implementations, a plurality of pillar interconnects (as described in FIG. **2**) may be used to couple the integrated devices to the substrate **102** and the bridge **130**. Stage 4 of FIG. **9**B illustrates and describes an example of coupling integrated devices to a substrate and a bridge.

The method couples (at **1025**) a bridge (e.g., **110**) to top portions of a plurality of integrated devices. For example, the bridge **110** may be coupled to a top portion of the integrated device **305** and a top portion of the integrated device **307**. The bridge **110** may be coupled to the integrated device **305** through the at least one solder interconnect **125**. The bridge **110** may be coupled to the integrated device **307** through the at least one solder interconnect **127**. A solder reflow process may be used to couple the bridge **110** to the integrated device **305** and the integrated device **307**. It is noted that in some implementations, a plurality of pillar interconnects (as described in FIG. **2**) may be used to couple the bridge **110** to the integrated device **305** and the integrated device **307**. Stage 5 of FIG. **9**C illustrates and describes an example of coupling a bridge to integrated devices.

The method couples (at **1030**) a plurality of solder interconnects (e.g., **124**) a bottom surface of the substrate (e.g., **102**). A solder reflow process may be used to couple the plurality of solder interconnects **124** to the substrate **102**. The plurality of solder interconnects **124** may be coupled to pad interconnects from the plurality of interconnects **122** of the substrate **102**. Stage 6 of FIG. **9**C illustrates and describes an example of solder interconnects coupled to substrate.

Exemplary Sequence for Fabricating a Substrate

In some implementations, fabricating a substrate includes several processes. FIGS. **11**A-**11**C illustrate an exemplary sequence for providing or fabricating a substrate. In some implementations, the sequence of FIGS. **11**A-**11**C may be used to provide or fabricate the substrate **202** and/or the substrate **204** of FIG. **2**. However, the process of FIGS. **11**A-**11**C may be used to fabricate any of the substrates described in the disclosure. In some implementations, at least some aspects of the process of FIGS. **11**A-**11**C may be used to fabricate the bridge **600**.

It should be noted that the sequence of FIGS. **11**A-**11**C may combine one or more stages in order to simplify and/or clarify the sequence for providing or fabricating a substrate. In some implementations, the order of the processes may be changed or modified. In some implementations, one or more of processes may be replaced or substituted without departing from the scope of the disclosure.

Stage 1, as shown in FIG. **11**A, illustrates a state after a carrier **1100** is provided and a metal layer is formed over the carrier **1100**. The metal layer may be patterned to form interconnects **1102**. A plating process and etching process may be used to form the metal layer and interconnects. In some implementations, the carrier **1100** may be provided with a metal layer that is patterned to form the interconnects **1102**.

Stage 2 illustrates a state after a dielectric layer **1120** is formed over the carrier **1100** and the interconnects **1102**. A deposition and/or lamination process may be used to form the dielectric layer **1120**. The dielectric layer **1120** may include polyimide. However, different implementations may use different materials for the dielectric layer.

Stage 3 illustrates a state after a plurality of cavities **1110** is formed in the dielectric layer **1120**. The plurality of cavities **1110** may be formed using an etching process (e.g., photo etching process) or laser process.

Stage 4 illustrates a state after interconnects **1112** are formed in and over the dielectric layer **1120**, including in and over the plurality of cavities **1110**. For example, a via, pad and/or traces may be formed. A plating process may be used to form the interconnects.

Stage 5 illustrates a state after another dielectric layer **1122** is formed over the dielectric layer **1120**. A deposition and/or lamination process may be used to form the dielectric layer **1122**. The dielectric layer **1122** may be the same material as the dielectric layer **1120**. However, different implementations may use different materials for the dielectric layer.

Stage 6, as shown in FIG. **11**B, illustrates a state after a plurality of cavities **1130** is formed in the dielectric layer **1122**. An etching process or laser process may be used to form the cavities **1130**.

Stage 7 illustrates a state after interconnects **1114** are formed in and over the dielectric layer **1122**, including in and over the plurality of cavities **1130**. For example, via, pad and/or trace may be formed. A plating process may be used to form the interconnects.

Stage 8 illustrates a state after another dielectric layer **1124** is formed over the dielectric layer **1122**. A deposition and/or lamination process may be used to form the dielectric layer **1124**. The dielectric layer **1124** may be the same material as the dielectric layer **1120**. However, different implementations may use different materials for the dielectric layer.

Stage 9 illustrates a state after a plurality of cavities **1140** is formed in the dielectric layer **1124**. An etching process or laser process may be used to form the cavities **1140**.

Stage 10, as shown in FIG. **11**C, illustrates a state after interconnects **1116** are formed in and over the dielectric layer **1124**, including in and over the plurality of cavities **1140**. For example, via, pad and/or trace may be formed. A plating process may be used to form the interconnects.

Some or all of the interconnects **1102**, **1112**, **1114** and/or **1116** may define a plurality of interconnects **122** of the substrate **102**. The dielectric layers **1120**, **1122**, **1124** may be represented by the at least one dielectric layer **120**.

Stage 11 illustrates a state after the carrier **1100** is decoupled (e.g., removed, grinded out) from the dielectric layer **1150**, leaving the substrate **102** that includes the at least one dielectric layer **120** and the plurality of interconnects **122**.

Stage 12 illustrates a state after the first solder resist layer **1160** and the second solder resist layer **1162** are formed over the substrate **102**. A deposition process may be used to form the first solder resist layer **1160** and the second solder resist layer **1162**. In some implementations, none or one solder resist layer may be formed over the at least one dielectric layer **1150**.

Different implementations may use different processes for forming the metal layer(s). In some implementations, a chemical vapor deposition (CVD) process and/or a physical vapor deposition (PVD) process for forming the metal layer(s). For example, a sputtering process, a spray coating process, and/or a plating process may be used to form the metal layer(s).

Exemplary Flow Diagram of a Method for Fabricating a Substrate

In some implementations, fabricating a substrate includes several processes. FIG. **12** illustrates an exemplary flow diagram of a method **1200** for providing or fabricating a substrate. In some implementations, the method **1200** of FIG. **12** may be used to provide or fabricate the substrate(s) of FIGS. **1**-**4**. For example, the method **1200** of FIG. **12** may be used to fabricate the substrate **102**.

It should be noted that the method of FIG. **12** may combine one or more processes in order to simplify and/or clarify the method for providing or fabricating a substrate. In some implementations, the order of the processes may be changed or modified.

The method provides (at **1205**) a carrier **1100**. Different implementations may use different materials for the carrier. The carrier may include a substrate, glass, quartz and/or carrier tape. Stage 1 of FIG. **11**A illustrates and describes an example of a carrier that is provided.

The method forms (at **1210**) a metal layer over the carrier **1100**. The metal layer may be patterned to form interconnects. A plating process may be used to form the metal layer and interconnects. In some implementations, the carrier may include a metal layer. The metal layer over the carrier may be patterned to form interconnects (e.g., **1102**). Stage 1 of FIG. **11**A illustrates and describes an example of a metal layer and interconnects that are formed over a carrier.

The method forms (at **1215**) a dielectric layer **1120** over the carrier **1100** and the interconnects **1102**. A deposition and/or lamination process may be used to form the dielectric layer. The dielectric layer **1120** may include polyimide. Forming the dielectric layer may also include forming a plurality of cavities (e.g., **1110**) in the dielectric layer **1120**. The plurality of cavities may be formed using an etching process (e.g., photo etching) or laser process. Stages 2-3 of FIG. **11**A illustrate and describe an example of forming a dielectric layer and cavities in the dielectric layer.

The method forms (at **1220**) interconnects in and over the dielectric layer. For example, the interconnects **1112** may be formed in and over the dielectric layer **1120**. A plating process may be used to form the interconnects. Forming interconnects may include providing a patterned metal layer over and/or in the dielectric layer. Forming interconnects may also include forming interconnects in cavities of the dielectric layer. Stage 4 of FIG. **11**A illustrates and describes an example of forming interconnects in and over a dielectric layer.

The method forms (at **1225**) a dielectric layer **1122** over the dielectric layer **1120** and the interconnects. A deposition and/or lamination process may be used to form the dielectric layer. The dielectric layer **1122** may include polyimide. Forming the dielectric layer may also include forming a plurality of cavities (e.g., **1130**) in the dielectric layer **1122**. The plurality of cavities may be formed using an etching process or laser process. Stages 5-6 of FIGS. **11**A-**11**B illustrate and describe an example of forming a dielectric layer and cavities in the dielectric layer.

The method forms (at **1230**) interconnects in and/or over the dielectric layer. For example, the interconnects **1114** may be formed. A plating process may be used to form the interconnects. Forming interconnects may include providing a patterned metal layer over an in the dielectric layer. Forming interconnects may also include forming interconnects in cavities of the dielectric layer. Stage 7 of FIG. **11**B illustrates and describes an example of forming interconnects in and over a dielectric layer.

The method may form additional dielectric layer(s) and additional interconnects as described at **1225** and **1230**. Stages 8-10 of FIGS. **11**B-**11**C illustrate and describe an example of forming an additional dielectric layer and interconnects in and over a dielectric layer.

Once all the dielectric layer(s) and additional interconnects are formed, the method may decouple (e.g., remove, grind out) the carrier (e.g., **1100**) from the dielectric layer

120, leaving the substrate. In some implementations, the method may form solder resist layers (e.g., **1160**, **1162**) over the substrate.

Different implementations may use different processes for forming the metal layer(s). In some implementations, a chemical vapor deposition (CVD) process and/or a physical vapor deposition (PVD) process for forming the metal layer(s). For example, a sputtering process, a spray coating process, and/or a plating process may be used to form the metal layer(s).

Exemplary Electronic Devices

FIG. **13** illustrates various electronic devices that may be integrated with any of the aforementioned device, integrated device, integrated circuit (IC) package, integrated circuit (IC) device, semiconductor device, integrated circuit, die, interposer, package, package-on-package (PoP), System in Package (SiP), or System on Chip (SoC). For example, a mobile phone device **1302**, a laptop computer device **1304**, a fixed location terminal device **1306**, a wearable device **1308**, or automotive vehicle **1310** may include a device **1300** as described herein. The device **1300** may be, for example, any of the devices and/or integrated circuit (IC) packages described herein. The devices **1302**, **1304**, **1306** and **1308** and the vehicle **1310** illustrated in FIG. **13** are merely exemplary. Other electronic devices may also feature the device **1300** including, but not limited to, a group of devices (e.g., electronic devices) that includes mobile devices, hand-held personal communication systems (PCS) units, portable data units such as personal digital assistants, global positioning system (GPS) enabled devices, navigation devices, set top boxes, music players, video players, entertainment units, fixed location data units such as meter reading equipment, communications devices, smartphones, tablet computers, computers, wearable devices (e.g., watches, glasses), Internet of things (IoT) devices, servers, routers, electronic devices implemented in automotive vehicles, or any other device that stores or retrieves data or computer instructions, or any combination thereof.

One or more of the components, processes, features, and/or functions illustrated in FIGS. **1**-**6**, **9**A-**9**C, **10**, **11**A-**11**C and/or **12**-**13** may be rearranged and/or combined into a single component, process, feature or function or embodied in several components, processes, or functions. Additional elements, components, processes, and/or functions may also be added without departing from the disclosure. It should also be noted FIGS. **1**-**6**, **9**A-**9**C, **10**, **11**A-**11**C and/or **12**-**13** and its corresponding description in the present disclosure is not limited to dies and/or ICs. In some implementations, FIGS. **1**-**6**, **9**A-**9**C, **10**, **11**A-**11**C and/or **12**-**13** and its corresponding description may be used to manufacture, create, provide, and/or produce devices and/or integrated devices. In some implementations, a device may include a die, an integrated device, an integrated passive device (IPD), a die package, an integrated circuit (IC) device, a device package, an integrated circuit (IC) package, a wafer, a semiconductor device, a package-on-package (PoP) device, a heat dissipating device and/or an interposer.

It is noted that the figures in the disclosure may represent actual representations and/or conceptual representations of various parts, components, objects, devices, packages, integrated devices, integrated circuits, and/or transistors. In some instances, the figures may not be to scale. In some instances, for purpose of clarity, not all components and/or parts may be shown. In some instances, the position, the location, the sizes, and/or the shapes of various parts and/or components in the figures may be exemplary. In some implementations, various components and/or parts in the figures may be optional.

The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any implementation or aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects of the disclosure. Likewise, the term "aspects" does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation. The term "coupled" is used herein to refer to the direct or indirect coupling (e.g., mechanical coupling) between two objects. For example, if object A physically touches object B, and object B touches object C, then objects A and C may still be considered coupled to one another—even if they do not directly physically touch each other. Coupling a component A to a component B may mean that a component A is coupled to a component B, and/or a component B is coupled to a component A. The term "electrically coupled" may mean that two objects are directly or indirectly coupled together such that an electrical current (e.g., signal, power, ground) may travel between the two objects. Two objects that are electrically coupled may or may not have an electrical current traveling between the two objects. The use of the terms "first", "second", "third" and "fourth" (and/or anything above fourth) is arbitrary. Any of the components described may be the first component, the second component, the third component or the fourth component. For example, a component that is referred to a second component, may be the first component, the second component, the third component or the fourth component. The term "encapsulating" means that the object may partially encapsulate or completely encapsulate another object. The terms "top" and "bottom" are arbitrary. A component that is located on top may be located over a component that is located on a bottom. A top component may be considered a bottom component, and vice versa. As described in the disclosure, a first component that is located "over" a second component may mean that the first component is located above or below the second component, depending on how a bottom or top is arbitrarily defined. In another example, a first component may be located over (e.g., above) a first surface of the second component, and a third component may be located over (e.g., below) a second surface of the second component, where the second surface is opposite to the first surface. It is further noted that the term "over" as used in the present application in the context of one component located over another component, may be used to mean a component that is on another component and/or in another component (e.g., on a surface of a component or embedded in a component). Thus, for example, a first component that is over the second component may mean that (1) the first component is over the second component, but not directly touching the second component, (2) the first component is on (e.g., on a surface of) the second component, and/or (3) the first component is in (e.g., embedded in) the second component. A first component that is located "in" a second component may be partially located in the second component or completely located in the second component. The term "about 'value X'", or "approximately value X", as used in the disclosure means within 10 percent of the 'value X'. For example, a value of about 1 or approximately 1, would mean a value in a range of 0.9-1.1.

In some implementations, an interconnect is an element or component of a device or package that allows or facilitates an electrical connection between two points, elements and/or components. In some implementations, an interconnect may include a trace, a via, a pad, a pillar, a redistribution metal layer, and/or an under bump metallization (UBM) layer. An interconnect may include one or more metal components (e.g., seed layer+metal layer). In some implementations, an interconnect is an electrically conductive material that may be configured to provide an electrical path for a signal (e.g., a data signal, ground or power). An interconnect may be part of a circuit. An interconnect may include more than one element or component. An interconnect may be defined by one or more interconnects. Different implementations may use similar or different processes to form the interconnects. In some implementations, a chemical vapor deposition (CVD) process and/or a physical vapor deposition (PVD) process for forming the interconnects. For example, a sputtering process, a spray coating, and/or a plating process may be used to form the interconnects.

Also, it is noted that various disclosures contained herein may be described as a process that is depicted as a flowchart, a flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed.

The following provides an overview of aspects of the present disclosure:

Aspect 1: A package comprising: a substrate; a first integrated device coupled to the substrate; a second integrated device coupled to the substrate; a first bridge coupled to the first integrated device and the second integrated device, wherein the first bridge is configured to provide at least one first electrical path between the first integrated device and the second integrated device, and wherein the first bridge is coupled to a top portion of the first integrated device and a top portion of the second integrated device; and a second bridge coupled to the first integrated device and the second integrated device, wherein the second bridge is configured to provide at least one second electrical path between the first integrated device and the second integrated device.

Aspect 2: The package of aspect 1, wherein the at least one first electrical path includes a back side of the first integrated device and a back side of the second integrated device.

Aspect 3: The package of aspects 1 through 2, wherein the first integrated device comprises: a first die; and a second die coupled to the first die.

Aspect 4: The package of aspect 3, wherein the second integrated device comprises: a third die; and a fourth die coupled to the third die.

Aspect 5: The package of aspect 4, wherein the first die is located over the second die, wherein the second die is coupled to the substrate, wherein the third die is located over the fourth die, and wherein the fourth die is coupled to the substrate.

Aspect 6: The package of aspects 4 through 5, wherein the first die is configured to be electrically coupled to the third die through the first bridge.

Aspect 7: The package of aspects 4 through 6, wherein the first integrated device includes a first encapsulation layer, and wherein the second integrated device includes a second encapsulation layer.

Aspect 8: The package of aspects 3 through 7, wherein a front side of the first die faces a front side of the second die.

Aspect 9: The package of aspects 3 through 7, wherein a front side of the first die faces a back side of the second die.

Aspect 10: The package of aspects 3 through 7, wherein a back side of the first die faces a front side of the second die.

Aspect 11: The package of aspects 1 through 10, wherein the first bridge includes a bridge die comprising: a die substrate; and at least one bridge interconnect.

Aspect 12: The package of aspects 1 through 10, wherein the first bridge includes a bridge structure comprising: at least one dielectric layer; and at least one bridge interconnect.

Aspect 13: The package of aspects 1 through 12, wherein the first bridge is configured to provide at least one first electrical path for input/output (I/O) signals between the first integrated device and the second integrated device.

Aspect 14: The package of aspects 1 through 13, wherein the second bridge is configured to provide at least one second electrical path for input/output (I/O) signals between the first integrated device and the second integrated device.

Aspect 15: The package of aspects 1 through 2, wherein the first integrated device comprises: a first die; and a second die coupled to the first die, wherein the second integrated device comprises: a third die; and a fourth die coupled to the third die, and wherein the first bridge is configured to provide at least one first electrical path for input/output (I/O) signals between the first die and the third die, and wherein the second bridge is configured to provide at least one second electrical path for input/output (I/O) signals between the second die and the fourth die.

Aspect 16: The package of aspects 1 through 15, wherein the second bridge is located at least partially in the substrate.

Aspect 17: An apparatus comprising: a substrate; a first integrated device coupled to the substrate; a second integrated device coupled to the substrate; means for a first bridge interconnection coupled to the first integrated device and the second integrated device, wherein the means for a first bridge interconnection is configured to provide at least one first electrical path between the first integrated device and the second integrated device, and wherein the means for a first bridge interconnection is coupled to a top portion of the first integrated device and a top portion of the second integrated device; and means for a second bridge interconnection coupled to the first integrated device and the second integrated device, wherein the means for a second bridge interconnection is configured to provide at least one second electrical path between the first integrated device and the second integrated device.

Aspect 18: The apparatus of aspect 17, wherein the first integrated device comprises: a first die; and a second die coupled to the first die.

Aspect 19: The apparatus of aspect 18, wherein the second integrated device comprises: a third die; and a fourth die coupled to the third die.

Aspect 20: The apparatus of aspects 18 through 19, wherein a front side of the first die faces a front side of the second die.

Aspect 21: The apparatus of aspects 18 through 19, wherein a front side of the first die faces a back side of the second die.

Aspect 22: The apparatus of aspects 18 through 19, wherein a back side of the first die faces a front side of the second die.

Aspect 23: The apparatus of aspects 17 through 22, wherein the means for a first bridge interconnection includes a bridge die comprising: a die substrate; and at least one bridge interconnect.

Aspect 24: The apparatus of aspects 17 through 22, wherein the means for a first bridge interconnection includes a bridge structure comprising: at least one dielectric layer; and at least one bridge interconnect.

Aspect 25: The apparatus of aspects 17 through 24, wherein the apparatus includes an electronic device selected from a group consisting of a music player, a video player, an entertainment unit, a navigation device, a communications device, a mobile device, a mobile phone, a smartphone, a personal digital assistant, a fixed location terminal, a tablet computer, a computer, a wearable device, a laptop computer, a server, an internet of things (IoT) device, and a device in an automotive vehicle.

Aspect 26: A method for fabricating a package, comprising: providing a substrate; coupling a first integrated device to the substrate; coupling a second integrated device to the substrate; coupling a first bridge to the first integrated device and the second integrated device, wherein the first bridge is configured to provide at least one first electrical path between the first integrated device and the second integrated device, and wherein the first bridge is coupled to a top portion of the first integrated device and a top portion of the second integrated device; and coupling a second bridge to the first integrated device and the second integrated device, wherein the second bridge is configured to provide at least one second electrical path between the first integrated device and the second integrated device.

Aspect 27: The method of aspect 26, wherein coupling the second bridge to the first integrated device and the second integrated device, is performed when the first integrated device and the second integrated device are coupled to the substrate.

Aspect 28: The method of aspects 26 through 27, wherein the at least one first electrical path includes a back side of the first integrated device and a back side of the second integrated device.

Aspect 29: The method of aspects 26 through 28, wherein the first integrated device comprises: a first die; a second die coupled to the first die; a third die; and a fourth die coupled to the third die.

Aspect 30: The method of aspect 29, wherein the first die is configured to be electrically coupled to the third die through the first bridge.

The various features of the disclosure described herein can be implemented in different systems without departing from the disclosure. It should be noted that the foregoing aspects of the disclosure are merely examples and are not to be construed as limiting the disclosure. The description of the aspects of the present disclosure is intended to be illustrative, and not to limit the scope of the claims. As such, the present teachings can be readily applied to other types of apparatuses and many alternatives, modifications, and variations will be apparent to those skilled in the art.

The invention claimed is:

1. A package comprising:

a first substrate comprising at least one dielectric layer;

a first integrated device coupled to the first substrate through at least a first plurality of pillar interconnects and a first plurality of solder interconnects;

a second integrated device coupled to the first substrate through at least a second plurality of pillar interconnects and a second plurality of solder interconnects;

a first bridge coupled to the first integrated device and the second integrated device through at least a third plurality of solder interconnects,

wherein the first bridge is configured to provide at least one first electrical path between the first integrated device and the second integrated device, and

wherein the first bridge is coupled to a top portion of the first integrated device and a top portion of the second integrated device, through at least the third plurality of solder interconnects; and

a second bridge coupled to the first integrated device and the second integrated device through a fourth plurality of solder interconnects,

wherein the second bridge is configured to provide at least one second electrical path between the first integrated device and the second integrated device,

wherein the second bridge comprises:

a silicon substrate comprising silicon;

a plurality of bridge interconnects;

a passivation layer coupled to the silicon substrate, wherein the passivation layer is a different material from the at least one dielectric layer of the first substrate;

wherein the fourth plurality of solder interconnects directly touches the plurality of bridge interconnects of the second bridge,

wherein the second bridge is embedded in the first substrate such that there is no dielectric layer from the first substrate that is located vertically between the passivation layer of the second bridge and the first integrated device, and

wherein a side portion of the passivation layer of the second bridge touches the first substrate.

2. The package of claim **1**, wherein the at least one first electrical path includes a back side of the first integrated device and a back side of the second integrated device.

3. The package of claim **1**, wherein the first integrated device comprises:

a first die; and

a second die coupled to the first die.

4. The package of claim **3**, wherein the second integrated device comprises:

a third die; and

a fourth die coupled to the third die.

5. The package of claim **4**,

wherein the first die is located over the second die,

wherein the second die is coupled to the first substrate,

wherein the third die is located over the fourth die, and

wherein the fourth die is coupled to the first substrate.

6. The package of claim **4**, wherein the first die is configured to be electrically coupled to the third die through the first bridge.

7. The package of claim **4**,

wherein the first integrated device includes a first encapsulation layer, and

wherein the second integrated device includes a second encapsulation layer.

8. The package of claim **3**, wherein a front side of the first die faces a front side of the second die.

9. The package of claim **3**, wherein a front side of the first die faces a back side of the second die.

10. The package of claim **3**, wherein a back side of the first die faces a front side of the second die.

11. The package of claim **1**, wherein the first bridge includes a bridge die comprising:

a first silicon substrate; and

at least one bridge interconnect.

12. The package of claim **1**, wherein the first bridge includes a bridge structure comprising:

at least one dielectric layer; and

at least one bridge interconnect.

13. The package of claim 1,
wherein the first bridge is configured to provide at least one first electrical path for input/output (I/O) signals between the first integrated device and the second integrated device, and
wherein the second bridge is configured to provide at least one second electrical path for input/output (I/O) signals between the first integrated device and the second integrated device.

14. The package of claim 1,
wherein the first integrated device is coupled to the second bridge through at least a third plurality of pillar interconnects and solder interconnects from the fourth plurality of solder interconnects, and
wherein the second integrated device is coupled to the second bridge through at least a fourth plurality of pillar interconnects and other solder interconnects from the fourth plurality of solder interconnects.

15. The package of claim 1, wherein the first bridge is coupled to the first integrated device and the second integrated device through at least a third plurality of pillar interconnects and the third plurality of solder interconnects.

16. The package of claim 1, wherein the second bridge is located at least partially in the first substrate.

17. A package comprising:
a first substrate comprising at least one dielectric layer;
a first stack of dies comprising:
a first die; and
a second die coupled to the first die through at least a first plurality of solder interconnects, wherein the second die is coupled to the first substrate through at least a first plurality of pillar interconnects and a second plurality of solder interconnects;
a second stack of dies comprising:
a third die; and
a fourth die coupled to the third die through at least a third plurality of solder interconnects, wherein the fourth die is coupled to the first substrate through at least a second plurality of pillar interconnects and a fourth plurality of solder interconnects;
a first bridge coupled to the first stack of dies and the second stack of dies through at least a fifth plurality of solder interconnects,
wherein the first bridge is configured to provide at least one first electrical path between the first stack of dies and the second stack of dies, and
wherein the first bridge is coupled to a top portion of the first stack of dies and a top portion of the second stack of dies, through at least the fifth plurality of solder interconnects; and
a second bridge coupled to the first stack of dies and the second stack of dies through a sixth plurality of solder interconnects,
wherein the second bridge is configured to provide at least one second electrical path between the first stack of dies and the second stack of dies,
wherein the second bridge comprises:
a silicon substrate comprising silicon;
a plurality of bridge interconnects;
a passivation layer coupled to the silicon substrate, wherein the passivation layer is a different material from the at least one dielectric layer of the first substrate;
wherein the sixth plurality of solder interconnects directly touches the plurality of bridge interconnects of the second bridge,
wherein the second bridge is embedded in the first substrate such that there is no dielectric layer from the first substrate that is located vertically between the passivation layer of the second bridge and the first stack of dies, and
wherein a side portion of the passivation layer of the second bridge touches the first substrate.

18. The package of claim 17, wherein a front side of the first die faces a front side of the second die.

19. The package of claim 17, wherein a front side of the first die faces a back side of the second die.

20. The package of claim 17, wherein a back side of the first die faces a front side of the second die.

21. The package of claim 17, wherein the first bridge includes a bridge die comprising:
a first silicon substrate; and
at least one bridge interconnect.

22. The package of claim 17, wherein the first bridge includes a bridge structure comprising:
at least one dielectric layer; and
at least one bridge interconnect.

23. The package of claim 17, wherein the second bridge is located at least partially in the first substrate.

24. The package of claim 17, wherein the first bridge is coupled to the first die and the third die through at least the fifth plurality of solder interconnects.

25. The package of claim 24, wherein the second bridge is coupled to the first second and the fourth die through at least the sixth plurality of solder interconnects.

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